

Spectral-truncation graph kernels for QAOA warm starts: topology-conditioned schedule transfer beyond depth one

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Training a variational quantum circuit is stochastic optimization over an objective a quantum device only ever samples, so the scarce resource is queries to that estimator and the effective lever is a good statistical prior over where the optimum lies. We build such a prior as a kernel over graphs: we introduce the spectral-truncation kernel (STK), a relabeling-invariant, positive-definite graph kernel that transfers an optimized QAOA angle schedule from the most similar training graph, and use it to overturn the prevailing depth-one verdict that learned warm starts only match physics-based initializers. The cost of the quantum approximate optimization algorithm is dominated by the circuit evaluations spent searching for good angles, and a topology-aware warm start removes that search only if it beats a simple ramp; prior comparisons, formed at circuit depth one, find no such margin. STK represents each graph by a finite low-frequency window of its normalized-Laplacian spectrum, the modes that fix the schedule, and donates the kernel-nearest graph’s depth- p schedule. On a query-counted MaxCut benchmark over 84 connected graphs from six families under family-held-out splits, STK reproduces the depth-one parity (paired advantage $+0.0000 \pm 0.0001$) and then beats the strongest adiabatic ramp by a margin that grows monotonically with depth, from $+0.0103$ at $p=2$ to $+0.0262$ at $p=3$, and by a wider one-shot margin still, $+0.0416$ at $p=2$ and $+0.0454$ at $p=3$, since the transferred schedule is near-optimal on the first query. Three deliverables make the claim checkable: proofs that the kernel is relabeling invariant and positive definite and that kernel schedule regression obeys a representer theorem, a truncation-error bound, and a generalization bound with linear-in-depth sample complexity; a depth-one objective cross-verified by two independent evaluators; and the pip-installable package `specops-stk`, whose single command `stk-reproduce` regenerates every figure, table, and number from fixed seeds.

I. INTRODUCTION

A quantum computer, approached from the keyboard, is a stochastic device: the Born rule returns one bit-string per run, so every quantity a variational algorithm cares about is a statistical estimate rather than a value read off directly. Training such an algorithm is thus stochastic optimization over an objective one can only estimate at nonzero variance and finite sample cost, and in that regime the classical statistical machinery of surrogate modeling, priors, and query-efficient design is not an add-on but the substance of the method. The expensive object here is the quantum expectation itself; a cheap, uncertainty-aware surrogate that predicts where its optimum sits, learned across related problem instances, is exactly what turns a blind search into a warm start. This work builds that surrogate as a kernel over graphs, integrating the domain structure of spectral graph theory so the learned prior is interpretable and provably invariant rather than a black box.

The quantum approximate optimization algorithm (QAOA) [1, 2] is a leading route to combinatorial optimization on near-term hardware, with a conjectured sampling hardness that motivates it as a candidate for early quantum advantage [40], and the workhorse benchmark for variational quantum algorithms in the noisy intermediate-scale quantum (NISQ) era [11, 12],

already executed on superconducting hardware at moderate scale [46]. Its target problem, MaxCut, is NP-hard [10] with a semidefinite-relaxation approximation as the classical bar [9], against which QAOA performance has been characterized numerically and analytically as a function of depth [41–43]. A depth- p QAOA circuit alternates a cost-Hamiltonian phase with a transverse-field mixer, itself a product-formula (Trotterized) circuit whose gate count and approximation error follow the operator-exponential structure analyzed for Hamiltonian simulation [7, 8, 56], and is parameterized by $2p$ angles (γ, β) that the algorithm tunes to maximize the expected cut $\langle C(\gamma, \beta) \rangle$. The bottleneck is the search itself, and it is statistical at root: each evaluation of $\langle C \rangle$ is an estimate assembled from many circuit executions whose sample complexity trades off against its variance, the classical optimizer calls this noisy estimator repeatedly, and the landscape is plagued by barren plateaus and cost concentration [4, 13, 14]. The operationally relevant figure of merit is therefore not the best achievable cut but the cut a policy reaches within a fixed query budget [57].

A warm start, which supplies good initial angles so the optimizer begins inside a high-quality basin, is the natural lever, and the QAOA landscape makes it viable: the objective concentrates across typical instances of a given structure [14], and optimal parameters transfer between random graphs and across problem sizes [15–17], a transferability that has been linked to graph similarity [47], to closed-form parameter-setting rules for weighted problems [48], and to the depth-asymptotic con-

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centration of QAOA angles on large-girth and mean-field instances [44]. Dedicated schemes place the optimizer in a good basin through continuation and annealing schedules [19], the interpolation (INTERP) growth of good angles with depth [4], classical-relaxation rounding [18], and classical-surrogate or meta-learning [20–22]. These invite a learning question that is squarely one of statistical surrogate modeling on structured objects: can a similarity model over graphs, fit on the optimized schedules of instances we have already solved, predict good angles for a graph we have not, and do so from a representation that respects the symmetries of the object? Concretely, can a policy that conditions on graph topology through a permutation-invariant representation, in the spirit of graph kernels [51–54] and of graph neural networks and Weisfeiler–Lehman kernels [24–29, 55] already productive for combinatorial optimization [30, 31], propose warm-start angles that transfer to unseen graph families and outperform simpler initializers? The proposals are many; controlled, leakage-aware, query-counted comparisons against a strong yet simple baseline are few. Where such controls have been applied, the depth-one verdict is negative: a learned policy only matches a one-line spectral or adiabatic rule, because at $p=1$ a single scalar already fixes the only relevant angle scale.

That verdict is an artifact of depth one, which we show by moving beyond it, where a warm start must specify an entire $2p$ -angle schedule rather than a single scale. Three observations organize the analysis. First, the depth-one parity is real but misleading: under a matched query budget every structure-aware policy ties an adiabatic ramp at $p=1$, which we reproduce exactly. Second, topology helps beyond depth one, because as p grows the instance-optimal schedule becomes a higher-dimensional object that a single physical scale can only crudely approximate, opening room for a fully topology-conditioned policy. Third, transferring a single optimized schedule is a more query-efficient form of that help than averaging. The QAOA landscape is multimodal: the optimal schedules of different graphs occupy distinct, symmetry-related basins, equally good optima separated by inferior regions. Copying one optimized schedule from a sufficiently similar graph lands the optimizer inside a genuine basin and is near-optimal on the first query, whereas regressing several schedules toward their mean places the seed between basins, a region the refiner must climb out of before it can improve. What remains is to identify which invariant notion of graph similarity governs schedule transfer.

We answer it with the spectral-truncation kernel (STK) (Fig. 1). Each graph is represented by a finite low-frequency window of its normalized-Laplacian spectrum, a spectral truncation of the graph operator that adapts the C^* -algebraic spectral-truncation kernels of Hashimoto *et al.* [5] from operators to graphs, and the kernel-nearest training graph donates its schedule. The truncated spectrum is a relabeling invariant, so the kernel is provably invariant and positive definite and the warm

start is a deterministic function of the isomorphism class. Our contributions are fourfold.

1. *Statistical theory.* We prove that the truncated spectrum and the full descriptor are relabeling invariant (Corollary 6, Theorem 7) and that STK is positive definite, making the warm start a deterministic function of the isomorphism class (Proposition 9, Corollary 10). Because the Born rule makes every landscape evaluation an estimate (Lemma 12), we then give the transfer its inferential footing: a representer theorem for kernel schedule regression (Theorem 13), a truncation-error bound for the rank- r spectral window (Proposition 14), a Rademacher generalization bound with sample complexity linear in depth (Theorem 15, Corollary 16), and a beyond-depth-one transfer guarantee (Theorem 18).
2. *Algorithm.* We introduce the STK warm-start operator, which transfers the optimized depth- p schedule of the kernel-nearest training graph; it is implemented, with all baselines and the exact evaluators, in the pip-installable package `specops-stk`, whose single command `stk-reproduce` regenerates every figure, table, and number in this paper from fixed seeds.
3. *Empirical finding.* Within a controlled and reproducible benchmark we show that the depth-one “learned warm starts only match” verdict is an artifact of $p=1$: topology-conditioned warm starts tie the ramp at $p=1$ ($+0.0000 \pm 0.0001$) and beat it for $p \geq 2$ by a margin that grows with depth, from $+0.0103$ at $p=2$ to $+0.0262$ at $p=3$, while STK transfer, near-optimal on the first query, beats the ramp by even more on the one-shot ratio than on the final ratio at every depth ($+0.0416$ versus $+0.0103$ at $p=2$, $+0.0454$ versus $+0.0262$ at $p=3$) and attains the best one-shot ratio of any policy through $p=2$; at $p=3$ the averaging descriptor-mean learner draws statistically level with transfer.
4. *Benchmark.* We supply the benchmark itself as a reusable yardstick: a proven relabeling-invariant descriptor and kernel, a depth-one objective cross-verified to 10^{-9} by two independent evaluators, exact approximation ratios against a brute-force Max-Cut oracle, and family-held-out transfer with a programmatic leakage check.

This work is the depth-scaling installment of a program that treats spectral truncation of an operator as a unifying primitive for query-efficient quantum optimization. The program develops graph-conditioned trust regions for query-efficient QAOA and their uncertainty calibration [57, 58], quantifies the measurement cost of warm-started low-depth QAOA and the query budgets needed to certify it [59, 60], and establishes topology-conditioned parameter transfer for budgeted graph optimization [61].

The operator-spectral truncated priors of [62] supply the descriptor most directly: the present spectral-truncation kernel makes that truncation the index of a relabeling-invariant similarity under which a single optimized schedule is transferred, and shows that the resulting warm start, null at $p=1$, separates from a single-scale ramp by a margin that grows with depth.

II. THE BUDGETED WARM-START BENCHMARK

A. QAOA objective and exact oracle

For MaxCut on $G = (V, E)$ the cost operator $C = \sum_{(u,v) \in E} \frac{1}{2}(1 - Z_u Z_v)$ counts the cut edges, rewarding configurations that disagree across an edge, and depth- p QAOA prepares a trial state by alternately phasing on this cost and mixing toward it,

$$|\gamma, \beta\rangle = \prod_{\ell=1}^p e^{-i\beta_\ell B} e^{-i\gamma_\ell C} |+\rangle^{\otimes n}, \quad (1)$$

starting from the uniform superposition $|+\rangle^{\otimes n}$ and driving transitions with the transverse-field mixer $B = \sum_v X_v$, the standard choice for unconstrained MaxCut among the broader family of QAOA mixers [45]; the algorithm then maximizes $\langle C(\gamma, \beta) \rangle$ over the $2p$ -angle schedule $\theta = (\gamma_1, \dots, \gamma_p, \beta_1, \dots, \beta_p)$. We evaluate $\langle C \rangle$ with an exact statevector simulator at cost $O(pn2^n)$ (Sec. V). At depth one we additionally compute the analytic per-edge closed form of Wang *et al.* [3] at cost $O(|E|)$; the two evaluators agree to 10^{-9} on every family, so the depth-one objective is analytically cross-verified and all reported objective values are exact. The quality of a cut is the approximation ratio $\langle C \rangle / C^*$ against the true MaxCut C^* , obtained by exact enumeration of the 2^{n-1} distinct bipartitions, which is tractable for $n \leq 16$. All ratios are ground truth rather than relaxation bounds.

B. Budgeted refiner and query-budget frontier

A warm-start policy maps a graph to an initial schedule θ_0 that seeds a coordinate-ascent refiner over all $2p$ angles. The refiner counts every call to the objective and stops when a per-graph budget of evaluations is exhausted, so the running-best ratio as a function of the number of evaluations, which we call the query-budget frontier, is a faithful and hardware-relevant comparison. Its first point, the one-shot ratio at $q=1$, measures pure warm-start quality with zero optimization, the regime that matters when each query is an expensive circuit execution.

C. Warm-start policies

We compare five policies. Two are non-learned adiabatic ramps [19] that can set only a single physical scale: *spectral* anchors a discretized annealing schedule to the Laplacian algebraic connectivity, and at $p=1$ reduces to the one-line rule $\gamma_0 = \pi/[2(1 + \lambda_2)]$, $\beta_0 = \pi/8$; *topology* anchors the same ramp to the mean degree. One policy is *random*. The remaining two are learned and predict the entire schedule from invariant topology: *descriptor mean*, a k -nearest-neighbor regressor ($k=3$) that averages the optimized schedules of the nearest training graphs, and *STK* (this work), which transfers the optimized schedule of the single training graph nearest under the spectral-truncation kernel. The transfer targets are near-optimal schedules from an INTERP-seeded optimizer [4] (Appendix B); they label training graphs only, and the learned policies must predict them on held-out families from topology alone.

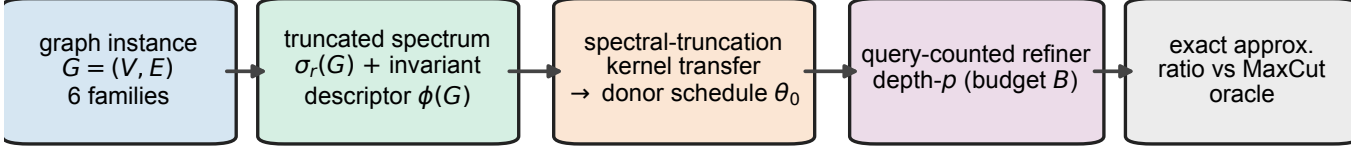
D. Families, splits, and metrics

The benchmark uses six structurally distinct families, generated with NetworkX [36] and reduced to the largest connected component: Erdős–Rényi with edge probability 0.5 [32], random 3-regular, Barabási–Albert (scale-free, $m=2$) [33], Watts–Strogatz (small-world, $k=4$, rewiring probability 0.3) [34], two-dimensional grid, and a two-block stochastic block model ($p_{\text{in}}=0.7$, $p_{\text{out}}=0.1$) [35]. Transfer is evaluated by family-held-out cross-validation: to evaluate a family, the learned policies are fit only on the union of the other five. A fingerprint built from family, n , $|E|$, and the hashed degree sequence confirms that no test graph appears in training, and the run records `leakage_clean=true`. Means are reported with 95% confidence intervals ($1.96s/\sqrt{n}$); advantages over baselines are paired across the held-out test graphs, and “best ramp” is the per-graph maximum of the spectral and topology ratios.

E. Reported scale and reproducibility

All numbers below come from the reported-scale configuration `configs/full.yaml`: six families with 84 connected graphs in total, sizes $n \in \{8, 10, 12, 14\}$, query budget 28, master seed 0, and depths $p \in \{1, 2, 3\}$. The leakage check confirms the run is clean (`leakage_clean=true`). The full multi-depth pipeline runs in 423.3409s with a peak memory of 163.2MB on a single laptop CPU. Every value is written to `results/summary.json` by the runner and propagated to the tables, figures, and the macros that typeset this manuscript by the plotting scripts, so no number is entered by hand.

family-held-out transfer (leakage-checked): schedules transferred, not averaged



kernel $k(G, G')$ positive-definite & relabeling-invariant • depth-1 closed-form $\leftrightarrow$ statevector to 10^{-9}

FIG. 1. Spectral-truncation-kernel schedule transfer for QAOA warm starts. A connected graph from one of six families is mapped to a relabeling-invariant representation: a truncated normalized-Laplacian spectrum $\sigma_r(G)$, a finite low-frequency window of the graph operator, together with an invariant topology descriptor $\phi(G)$. A positive-definite spectral-truncation kernel $k(G, G')$ selects the single most similar training graph and copies its optimized depth- p angle schedule θ_0 , transferring one real optimum rather than averaging several. The schedule seeds a refiner that counts every objective evaluation against a fixed budget B ; quality is the exact approximation ratio against a brute-force MaxCut oracle. Two guarantees underpin the benchmark: the kernel is positive definite and invariant, $k(\pi G, \cdot) = k(G, \cdot)$, and the depth-one objective is computed in two independent ways, an analytic closed form and an exact statevector simulation, agreeing to 10^{-9} . Policies are compared under family-held-out splits with a programmatic leakage check.

III. THE SPECTRAL-TRUNCATION KERNEL

At depth one the closed-form objective depends only on local degree and triangle structure, quantities that a single Laplacian scale already captures, which is why no policy beats the ramp there. At larger depth the optimal schedule is a higher-dimensional object whose shape is set by the low-frequency structure of the instance, including community structure, regularity, and connectivity bottlenecks. That information is carried by the smallest non-zero eigenvalues of the normalized Laplacian, the same low-frequency modes that govern connectivity, mixing, and community structure in spectral graph theory [49] and that drive spectral clustering and partitioning [50]. Truncating the spectrum to this low-frequency window yields an invariant fingerprint under which “similar graph” means “similar optimal schedule.”

Each graph G is mapped to two relabeling-invariant objects. The topology descriptor $\phi(G) \in \mathbb{R}^m$ concatenates degree statistics, motif densities from traces of powers of the adjacency matrix, a hashed Weisfeiler-Lehman colour-refinement histogram [24, 25], cycle features, Laplacian-spectral features, and connectivity features (Sec. V). The spectral-truncation feature $\sigma_r(G)$ retains the $r = 6$ smallest non-zero eigenvalues of the normalized Laplacian $\mathcal{L} = D^{-1/2}LD^{-1/2}$, in increasing order and right-padded to length r ; these low-lying modes encode the global organization of the graph, its community structure, regularity, and connectivity bottlenecks, while discarding the high-frequency detail that does not shape the optimal schedule. Both are invariant under node relabeling, as established below.

On standardized features, the spectral-truncation ker-

nel measures graph similarity as a product of two Gaussian factors, one comparing low-frequency spectra and one comparing the topology descriptor, so that two graphs score as similar only when they agree on both,

$$k(G, G') = \exp\left(-\frac{\|\sigma_r(G) - \sigma_r(G')\|^2}{2\ell_s^2}\right) \times \exp\left(-\frac{\|\phi(G) - \phi(G')\|^2}{2\ell_d^2}\right), \quad (2)$$

with bandwidths $\ell_s, \ell_d > 0$ set by the median heuristic. The STK policy outputs $\Theta(G) = \theta_{j^*}$ with $j^* = \arg \max_i k(G, G_i)$, the optimized schedule of the kernel-nearest training graph. The two feature maps are relabeling invariant, so k is invariant and, as a Hadamard product of Gaussian radial basis functions, positive definite; the transfer policy is consequently a deterministic function of the isomorphism class. This places STK on the same footing as the established operator-truncation kernels of Hashimoto *et al.* [5], with the graph Laplacian as the truncated operator and QAOA schedule transfer as the downstream task. A kernel-ridge variant that averages over donors via $(K + \lambda I)^{-1}$ is included in the package as an ablation and reproduces the between-basins behavior of the descriptor mean.

IV. RESULTS

A. The warm-start advantage grows with depth

Fig. 2 and Table I report the held-out approximation ratio as a function of QAOA depth, and the advantage of learned transfer is null at depth one and grows monotonically thereafter. At $p=1$ all structure-aware policies are

statistically indistinguishable: STK, the descriptor mean, and both ramps land within statistical resolution of one another, with STK at 0.7825 and a paired STK–ramp difference of $+0.0000 \pm 0.0001$, consistent with zero. This reproduces the established depth-one parity. When the warm start is a single (γ, β) pair, one scalar already fixes the only relevant angle scale: the Laplacian connectivity used by the ramp and the topology seen by the learned policy carry the same single-scale signal, and a modest refinement budget erases any head start.

The picture changes qualitatively for $p \geq 2$, where the warm start must specify a $2p$ -angle schedule that a single physical scale can only crudely approximate. STK exceeds the strongest per-instance ramp by +0.0103 on the final ratio and by +0.0416 on the one-shot ratio at $p=2$, and by +0.0262 (final) and +0.0454 (one-shot) at $p=3$. Both final advantages exceed their paired 95% confidence intervals, ± 0.0019 at $p=2$ and ± 0.0026 at $p=3$, and at each depth the one-shot advantage, evaluated with zero optimization, exceeds the final advantage, +0.0416 versus +0.0103 at $p=2$ and +0.0454 versus +0.0262 at $p=3$. In absolute terms STK reaches 0.8503 at $p=2$ and 0.8902 at $p=3$ against oracle ceilings of 0.8536 and 0.8999, closing most of the gap the ramp leaves open. The trend is monotone in depth: the advantage is null at $p=1$ and grows as the schedule becomes higher-dimensional and the single-scale ramp becomes a coarser approximation to the instance-optimal schedule. The averaging-based descriptor mean also improves on the ramps at depth, which confirms that conditioning on topology, rather than the specific estimator, is what helps; at the lower depths it does so with far less query efficiency than transfer, as we show next.

B. Per-policy performance and query efficiency

Table II reports, at the primary depth $p=2$, the held-out final approximation ratio (mean over the $n = 84$ test evaluations with 95% confidence interval), the one-shot ($q=1$) ratio, the mean queries to a target of 0.95, and the target hit rate. STK attains the best one-shot ratio (0.8449) by a decisive margin and the highest final ratio (0.8503), narrowly exceeding the descriptor mean (0.8470) on the final ratio within overlapping confidence intervals: its transferred schedule is already near its final quality on the first evaluation, whereas the ramps start far lower (0.8032 for topology, 0.7483 for spectral) and the random control lower still (0.6057). This one-shot gap is the principal separation between the two learned policies at this depth. The descriptor mean reaches a competitive final ratio (0.8470, itself above both ramps) but a much weaker one-shot ratio (0.8050), because its between-basins average must be repaired by the refiner, whereas the single transferred optimum of STK needs no repair. Transfer is therefore the more query-efficient form of the same topology conditioning, delivering near-final quality per query. No policy reached the stringent

0.95 target within 28 queries at this depth, so queries-to-target equals the budget and the hit rate is zero for all policies; we report this transparently rather than relaxing the target.

C. Query-budget frontier

Fig. 3 plots the mean running-best approximation ratio against the query budget, one panel per depth. At $p=1$ the structure-aware curves are superimposed and the refiner equalizes them within a few queries. At $p=2$ and $p=3$ the STK curve starts far above the ramps, with its one-shot value at $p=2$ (0.8449) already exceeding their final values (0.8394 and 0.8319), and stays above them throughout: the local refiner improves all policies but cannot move a poorly seeded ramp out of the inferior basin it started in, so the head start is not equalized as it was at depth one. The two learned policies differ most at $p=2$, where the averaging-based learner starts near the ramps and well below STK, consistent with a between-basins seed that the refiner must repair, and closes much of the gap only by the end of the budget; at $p=3$ the averaging learner already begins near STK, so the early-budget gap between the two learned policies has largely closed. Table IV reports the frontier at selected query indices.

D. Per-family transfer

Fig. 4 breaks the held-out ratio at $p=2$ down by graph family. The transfer advantage is broad rather than driven by a single family: STK matches or exceeds the ramps across structurally diverse families, including community-structured stochastic-block and Erdős–Rényi graphs, scale-free Barabási–Albert graphs, small-world Watts–Strogatz graphs, and locally lattice-like grids, with the largest gains where the instance-optimal schedule departs most from a single-scale ramp. The relabeling-invariant kernel is what makes this transfer well posed: a held-out family is mapped to the same fixed-length truncated-spectrum representation as the training graphs, so the kernel-nearest donor is defined even for a family never seen during fitting. Table III reports the per-family numbers.

E. Cross-verification and invariance

Two internal correctness checks underpin the benchmark. The depth-one objective is computed by the analytic per-edge closed form [3] and independently by the exact statevector simulator; across all six families and a panel of angle settings the two agree to better than 10^{-9} , so the statevector evaluator used at every depth is provably the same objective as the analytic one where the

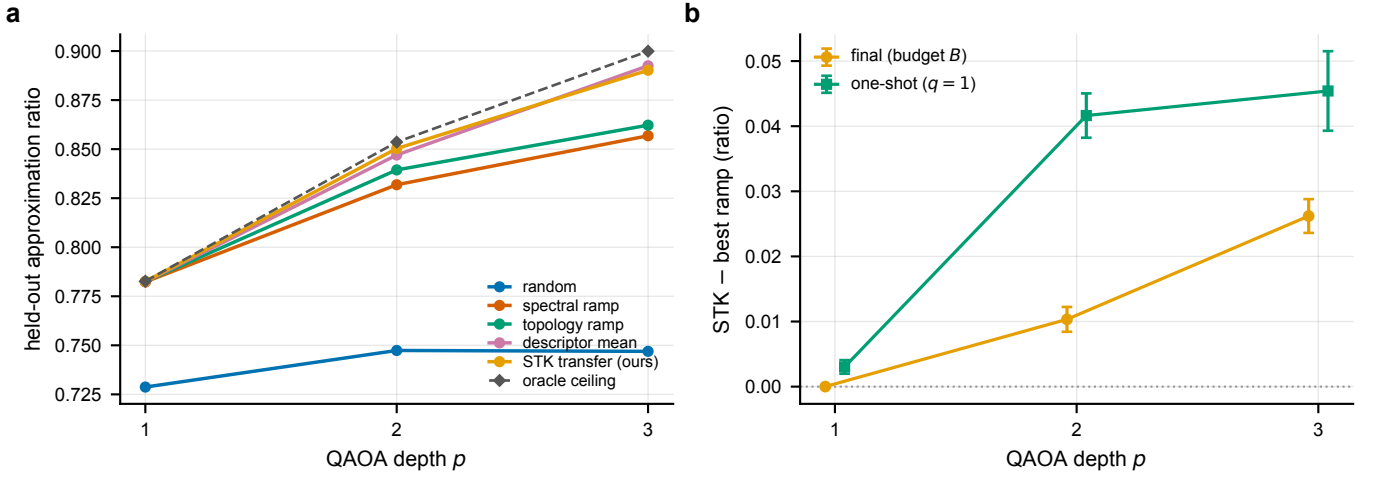


FIG. 2. The learned-transfer advantage over an adiabatic ramp is null at depth one and grows with QAOA depth. (a) Mean held-out approximation ratio versus depth p for each policy, with the exact-optimum oracle ceiling (dashed): the structure-aware policies coincide at $p=1$, while the spectral-truncation-kernel transfer policy (STK) separates upward from the adiabatic ramps and from the averaging-based descriptor mean as p grows, tracking the oracle. (b) The paired advantage of STK over the best per-instance ramp, both final (at the full budget B) and one-shot ($q=1$), with 95% confidence intervals; it is consistent with zero at $p=1$ and rises to $+0.0103$ at $p=2$ and $+0.0262$ at $p=3$ on the final ratio, outside its paired confidence interval. Lines and markers are means over the $n = 84$ held-out test graphs; error bars are 95% confidence intervals of the paired mean ($1.96 s/\sqrt{n}$). Data: `advantage_vs_depth` in `results/summary.json`.

TABLE I. Held-out approximation ratio versus QAOA depth. For each depth we report the best per-instance adiabatic ramp, the averaging-based descriptor mean, the spectral-truncation-kernel transfer policy (STK, this work), the exact-optimum oracle ceiling, and the paired STK–ramp advantage on the final and one-shot ratios (Δ_{final} with its 95% confidence interval, and $\Delta_{\text{one-shot}}$). STK ties the ramp at $p=1$ and beats it by a margin that grows with depth; the descriptor mean also improves on the ramp at $p \geq 2$ but with weaker one-shot quality. Bold marks STK where it meets or exceeds the ramp. Values are generated by `scripts/make_tables.py` from `results/summary.json`.

p	Best ramp	Descriptor mean	STK (ours)	Oracle	Δ_{final}	$\Delta_{\text{one-shot}}$
1	0.7825	0.7824	0.7825	0.7827	$+0.0000 \pm 0.0001$	$+0.0030$
2	0.8394	0.8470	0.8503	0.8536	$+0.0103 \pm 0.0019$	$+0.0416$
3	0.8622	0.8925	0.8902	0.8999	$+0.0262 \pm 0.0026$	$+0.0454$

latter applies. Separately, the truncated-spectrum representation and the full topology descriptor are verified invariant to node relabeling: under repeated random permutations they are unchanged to 10^{-8} , and the resulting kernel Gram matrix is symmetric and positive semidefinite. These guarantees ensure that the reported ratios are exact and that the learned policy sees only order-invariant graph information. The next section places each guarantee on a rigorous footing.

V. THEORETICAL GUARANTEES

This section formalizes and proves the structural guarantees on which the benchmark rests. We show that the exact oracle is correct and the global $\mathbb{Z}_2$ symmetry may be quotiented (Lemma 1); that the transverse-field mixer factorizes, so the simulator is matrix-free (Lemma 2); that the depth-one expectation admits a per-edge closed form and equals the statevector expectation

exactly (Proposition 3 and Corollary 4); that traces of adjacency powers, and hence the truncated Laplacian spectrum, are relabeling invariant (Lemma 5 and Corollary 6); that the full descriptor map is relabeling invariant (Theorem 7); that Weisfeiler–Lehman refinement is permutation equivariant and upper-bounds message-passing expressivity (Proposition 8); that the spectral-truncation kernel of Eq. (2) is positive definite and invariant and the transfer policy is therefore a deterministic class function (Proposition 9 and Corollary 10); and that the running-best curve of the refiner is monotone and pinned at its warm start, which, together with basin locality, is why a depth- p head start is not equalized (Proposition 11).

A. Notation and standing assumptions

Throughout, $G = (V, E)$ is a finite simple undirected graph on $n = |V|$ vertices with adjacency matrix $A \in \{0, 1\}^{n \times n}$ (symmetric, zero diagonal), degree

TABLE II. Per-policy warm-start benchmark at the primary depth $p=2$. Mean held-out depth-2 approximation ratio over $n = 84$ family-held-out test evaluations ($\pm 95\%$ confidence interval, $1.96 s/\sqrt{n}$); one-shot ($q=1$) ratio; mean queries to reach the 0.95 target (capped at the budget $B = 28$); and target hit rate. STK attains the best final and one-shot ratios; the descriptor mean reaches a competitive final ratio but a much weaker one-shot ratio. Bold marks the highest value in the final-ratio column. Values transcribed by code from `results/summary.json`.

Policy	Approx. ratio (final)	One-shot ($q=1$)	Queries $\rightarrow$ target	Hit rate
Random	0.7474 ± 0.0208	0.6057	28.00	0.00
Spectral ramp	0.8319 ± 0.0095	0.7483	28.00	0.00
Topology ramp	0.8394 ± 0.0088	0.8032	28.00	0.00
Descriptor mean	0.8470 ± 0.0080	0.8050	28.00	0.00
STK transfer (ours)	0.8503 ± 0.0081	0.8449	28.00	0.00

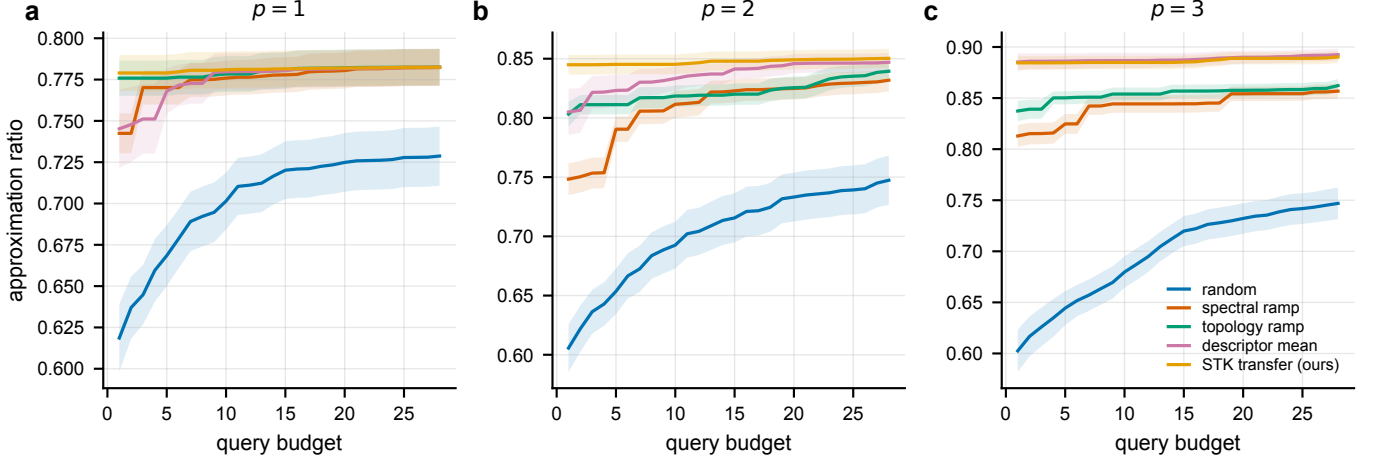


FIG. 3. The query-budget frontier shows that a higher-quality warm start is not equalized by refinement beyond depth one. Each panel plots the mean running-best held-out approximation ratio versus the number of objective evaluations, one per depth p in (a)–(c). The STK transfer policy begins above the adiabatic ramps at $p \geq 2$ and remains above them across the budget; at $p=1$ all structure-aware curves coincide. Lines are means over the $n = 84$ held-out graphs; shaded bands are 95% confidence intervals of the mean ($1.96 s/\sqrt{n}$). Data: the `frontier` blocks of `results/summary.json`.

matrix $D = \text{diag}(d_1, \dots, d_n)$, combinatorial Laplacian $L = D - A$, and normalized Laplacian $\mathcal{L} = D^{-1/2} L D^{-1/2}$. A node relabeling is a permutation $\pi \in S_n$ with permutation matrix Π ; the relabeled graph πG has adjacency $A(\pi G) = \Pi A \Pi^\top$. We write $|z\rangle$ for the computational basis state indexed by $z \in \{+1, -1\}^n$, $Z_v |z\rangle = z_v |z\rangle$, and X_v for the bit flip on qubit v . The cost operator is $C = \sum_{(u,v) \in E} \frac{1}{2} (1 - Z_u Z_v)$, the mixer is $B = \sum_v X_v$, and the depth- p state is given by Eq. (1). The exact oracle is invoked only for $n \leq 20$, so brute-force enumeration is well defined.

B. Correctness of the exact MaxCut oracle

Lemma 1 ($\mathbb{Z}_2$ reduction and oracle correctness). *The cut value $C(z) = \sum_{(u,v) \in E} \frac{1}{2} (1 - z_u z_v)$ satisfies $C(-z) = C(z)$ for all $z \in \{+1, -1\}^n$. Consequently, fixing $z_0 = +1$ and enumerating the 2^{n-1} configurations of $(z_1, \dots, z_{n-1})$ visits every cut exactly once up to a global flip, and the maximum so obtained equals the true MaxCut value $C^* = \max_z C(z)$.*

Proof. For any z , $(-z)_u (-z)_v = z_u z_v$, so each summand $\frac{1}{2} (1 - z_u z_v)$ is unchanged and $C(-z) = C(z)$; thus C is constant on each orbit $\{z, -z\}$ of the global sign-flip action of $\mathbb{Z}_2$. The action is free (since $z \neq -z$), so the 2^n configurations partition into 2^{n-1} orbits, each with a unique representative having $z_0 = +1$. Enumerating those representatives evaluates C once per orbit, and since C is orbit-constant, $\max\{C(z) : z_0 = +1\} = C^*$. The enumeration is exhaustive and exact. $\square$

This certifies that every approximation ratio $\langle C \rangle / C^*$ in this work is measured against ground truth rather than a relaxation bound.

C. Matrix-free mixer and the depth-one closed form

Lemma 2 (Factorization of the mixer unitary). *The mixer $B = \sum_v X_v$ exponentiates as a tensor product of*

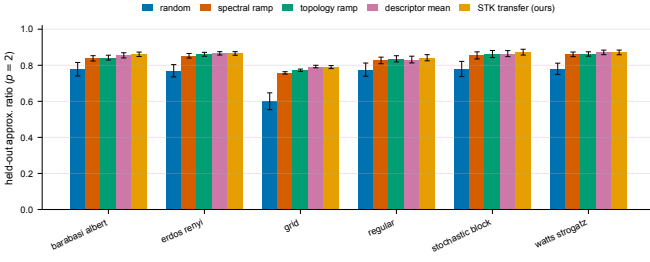


FIG. 4. The depth-2 transfer advantage is broad across graph families rather than driven by any single one. Grouped bars show the mean held-out approximation ratio for each policy within each of the six families; for each family the learned policies are fit only on the other five. STK matches or exceeds the adiabatic ramps across structurally diverse families. Bars are means and error bars are 95% confidence intervals of the mean ($1.96 s/\sqrt{n}$) over the held-out graphs in each family. Data: the primary-depth by_family block of `results/summary.json`.

single-qubit rotations,

$$e^{-i\beta B} = \bigotimes_{v=1}^n e^{-i\beta X_v} = \bigotimes_{v=1}^n (\cos \beta \mathbb{I} - i \sin \beta X), \quad (3)$$

so $e^{-i\beta B}$ acts by n independent 2×2 operations and is never assembled as a $2^n \times 2^n$ matrix.

Proof. The $\{X_v\}_v$ act on distinct tensor factors and commute pairwise, so $e^{-i\beta \sum_v X_v} = \prod_v e^{-i\beta X_v} = \bigotimes_v e^{-i\beta X_v}$. Since $X_v^2 = \mathbb{I}$, the power series splits into even and odd parts, $e^{-i\beta X_v} = \cos \beta \mathbb{I} - i \sin \beta X_v$, which gives Eq. (3). Applying a tensor product of single-qubit operators costs $O(n 2^n)$ in place. $\square$

This makes the statevector evaluator cost $O(p n 2^n)$ rather than $O(2^{2n})$ at every depth, and underwrites the depth- p objective used throughout.

Proposition 3 (Depth-one per-edge closed form). *Let $|\gamma, \beta\rangle = e^{-i\beta B} e^{-i\gamma C} |+\rangle^{\otimes n}$. For an edge $(u, v) \in E$ write $\langle C_{uv} \rangle = \langle \gamma, \beta | \frac{1}{2}(1 - Z_u Z_v) | \gamma, \beta \rangle$, let d_u, d_v be the numbers of neighbours of u, v other than the partner, and f the number of common neighbours. Then*

$$\langle C_{uv} \rangle = \frac{1}{2} + \frac{1}{4} \sin(4\beta) \sin \gamma [\cos^{d_u} \gamma + \cos^{d_v} \gamma] - \frac{1}{4} \sin^2(2\beta) \cos^{d_u+d_v-2f} \gamma [1 - \cos^f(2\gamma)], \quad (4)$$

and $\langle C \rangle = \sum_{(u,v) \in E} \langle C_{uv} \rangle$ by linearity.

Proof. By linearity it suffices to evaluate a single edge. Conjugating through the unitaries, $\langle C_{uv} \rangle = \langle + |^{\otimes n} U^\dagger \frac{1}{2}(1 - Z_u Z_v) U | + \rangle^{\otimes n}$ with $U = e^{-i\beta B} e^{-i\gamma C}$. The mixer conjugation rotates each Pauli on a qubit w in the Y - Z plane, $e^{i\beta B} Z_w e^{-i\beta B} = \cos(2\beta) Z_w - \sin(2\beta) Y_w$, which factorizes over qubits by Lemma 2 (using $XZX = -Z$, $XYX = -Y$). Hence $e^{i\beta B} Z_u Z_v e^{-i\beta B}$ expands into four terms with coefficients $\cos^2 2\beta$, $-\sin 2\beta \cos 2\beta$

(twice), and $\sin^2 2\beta$. Taking expectations in $e^{-i\gamma C} |+\rangle^{\otimes n}$, where C couples only edges so each expectation factorizes over the local neighbourhood, gives a vanishing $Z_u Z_v$ term; the cross terms $Y_u Z_v$ and $Z_u Y_v$ each contribute $\frac{1}{2} \sin(2\beta) \cos(2\beta) \sin \gamma \cos^{d_u} \gamma$ (resp. d_v); and $Y_u Y_v$ contributes $-\frac{1}{4} \sin^2(2\beta) \cos^{d_u+d_v-2f} \gamma [1 - \cos^f(2\gamma)]$. Collecting and using $2 \sin(2\beta) \cos(2\beta) = \sin(4\beta)$ yields Eq. (4), which is the result of Wang *et al.* [3]. $\square$

Corollary 4 (Closed-form/statevector equivalence). *For every (γ, β) and every G , the per-edge closed form of Proposition 3 summed over E equals the statevector expectation $\langle \gamma, \beta | C | \gamma, \beta \rangle$ of Lemma 2. The two are the same real-analytic function, so any numerical discrepancy is floating-point only.*

Proof. Both equal $\langle \gamma, \beta | C | \gamma, \beta \rangle$: the statevector evaluator computes it by exact application of $e^{-i\gamma C}$ and the factorized $e^{-i\beta B}$ followed by the diagonal C , while Proposition 3 derives the same expectation in closed form. Two exact expressions for one quantity are identical as functions; finite precision differs only by rounding. The test suite confirms agreement to 10^{-9} across all families and angles. $\square$

Corollary 4 licenses the depth-one cross-verification reported in Sec. IV and certifies the statevector objective used at $p \geq 2$ against the analytic one at $p=1$.

D. Relabeling invariance of the representation

Lemma 5 (Invariance of adjacency-power traces). *For every integer $k \geq 0$ and every permutation matrix Π , $\text{tr}((\Pi A \Pi^\top)^k) = \text{tr}(A^k)$. Consequently the triangle count $\frac{1}{6} \text{tr}(A^3)$, the closed-4-walk count $\text{tr}(A^4)$, the degree multiset, and every Laplacian eigenvalue are invariant under node relabeling.*

Proof. Since Π is orthogonal, $(\Pi A \Pi^\top)^k = \Pi A^k \Pi^\top$ and $\text{tr}(\Pi A^k \Pi^\top) = \text{tr}(A^k)$. The motif counts are fixed multiples of these traces; the degree multiset is permuted but unchanged as a multiset; and $L(\pi G) = \Pi L \Pi^\top$, $\mathcal{L}(\pi G) = \Pi \mathcal{L} \Pi^\top$ are similar to $L, \mathcal{L}$, hence isospectral. $\square$

Corollary 6 (Invariance of the truncated spectrum). *For every r , the spectral-truncation feature $\sigma_r(G)$, the r smallest non-zero eigenvalues of $\mathcal{L}$ in increasing order right-padded to length r , satisfies $\sigma_r(\pi G) = \sigma_r(G)$ for all $\pi \in S_n$.*

Proof. By Lemma 5 the spectrum of $\mathcal{L}$ is invariant; selecting the r smallest non-zero eigenvalues in increasing order and padding is a fixed function of the ordered spectrum, so σ_r is invariant. $\square$

Theorem 7 (Relabeling invariance of the descriptor map). *Let $\phi(G) \in \mathbb{R}^m$ be the topology descriptor formed by concatenating the degree, motif, hashed*

Weisfeiler–Lehman, cycle, Laplacian-spectral, and connectivity blocks of Sec. III. Then $\phi(\pi G) = \phi(G)$ for every $\pi \in S_n$.

Proof. Each block is invariant, so their concatenation is. The degree block consists of symmetric functions of the invariant degree multiset. The motif block uses normalized $\text{tr}(A^3)$ and $\text{tr}(A^4)$ and the average clustering, all invariant by Lemma 5. For the WL block, the colour multiset is permutation invariant by Proposition 8, and the hashed histogram bins it. The cycle block uses fractions defined by subgraph incidence, normalized by n . The Laplacian block uses symmetric functions of the invariant L - and $\mathcal{L}$ -spectra. The connectivity block uses assortativity and global clustering, built from invariant edge- and degree-multisets, with non-finite intermediates mapped to a constant 0. $\square$

The test suite verifies Theorem 7 and Corollary 6 numerically to 10^{-8} over random permutations.

E. Weisfeiler–Lehman equivariance and expressivity

Proposition 8 (WL permutation equivariance and message-passing upper bound). *Let $c_v^{(t)}$ be the Weisfeiler–Lehman colour of v after t rounds, initialized by degree and updated by $c_v^{(t+1)} = \text{hash}(c_v^{(t)}, \{\!\!\{ c_u^{(t)} : u \in N(v) \}\!\!\})$. Then (i) $c_{\pi(v)}^{(t)}(\pi G) = c_v^{(t)}(G)$ for all t, v , so the colour multiset and its histogram are permutation invariant; and (ii) any message-passing graph neural network with permutation-invariant readout and the same initial features cannot distinguish two graphs that WL fails to distinguish.*

Proof. (i) Induct on t . At $t = 0$ the colour is the degree, and $d_{\pi(v)}(\pi G) = d_v(G)$. Assuming the claim at t , the neighbour-colour multiset of $\pi(v)$ in πG equals that of v in G , because a multiset ignores labels, so the same hash gives equal colours at $t + 1$. (ii) A message-passing layer $h_v^{(t+1)} = \psi(h_v^{(t)}, \bigoplus_{u \in N(v)} \varphi(h_u^{(t)}))$ with invariant $\bigoplus$ refines no finer than WL by a parallel induction, so the graph-level readout is constant on any pair WL identifies [24, 28]. $\square$

Proposition 8 justifies the WL histogram as a principled, dependency-light stand-in for a graph-neural-network block of the descriptor.

F. The spectral-truncation kernel and the transfer policy

Proposition 9 (The spectral-truncation kernel is invariant and positive definite). *Let $k(G, G')$ be the kernel of Eq. (2) on standardized features, with bandwidths $\ell_s, \ell_d > 0$. Then (i) $k(\pi G, G') = k(G, G')$ and $k(G, \pi G') =$*

$k(G, G')$ for all permutations; and (ii) k is positive definite, so for any finite set of graphs the Gram matrix $K_{ij} = k(G_i, G_j)$ is symmetric positive semidefinite.

Proof. (i) Both feature maps are relabeling invariant, σ_r by Corollary 6 and ϕ by Theorem 7; the standardization is a fixed affine map, and k depends on G only through them, so it is invariant. (ii) Each factor is a Gaussian radial basis function on a Euclidean feature space, which is positive definite [6]. The Hadamard product of two positive-definite kernels is positive definite by the Schur product theorem, so k is positive definite. $\square$

Corollary 10 (STK transfer is a deterministic class function). *Fix a training set of (graph, optimized-schedule) pairs $\{(G_i, \theta_i)\}$ and the standardization estimated on it. For a test graph G , the STK policy outputs $\Theta(G) = \theta_{j^*}$ with $j^* = \arg \max_i k(G, G_i)$. Then $\Theta(\pi G) = \Theta(G)$ for every π , and Θ is a deterministic function of the isomorphism class of G given the seeded training set; ties are broken by a labeling-independent rule on training indices.*

Proof. By Proposition 9(i) the kernel values $\{k(G, G_i)\}_i$ are unchanged under relabeling of G , so the arg max index j^* and hence θ_{j^*} are unchanged; given the fixed training set the computation has no randomness, so Θ is reproducible. $\square$

Corollary 10 is the formal sense in which STK is a principled, invariant estimator rather than a label-dependent lookup, and is what makes the family-held-out transfer of Fig. 4 meaningful.

G. Monotone refinement, the warm-start floor, and basin locality

Proposition 11 (Monotonicity, warm-start floor, and non-equalization). *Let r_t be the running-best approximation ratio after t queries of the coordinate-ascent refiner started from a schedule with ratio r_0 . Then (i) r_t is monotone non-decreasing and (ii) $r_t \geq r_0$ for all t . Consequently (iii) if two policies seed the refiner in different basins of attraction of the coordinate-ascent dynamics, and the budget is exhausted before either escapes its basin, then the policy with the higher-quality basin retains its advantage: its r_t cannot be overtaken by a policy trapped below it.*

Proof. The refiner replaces the incumbent only on strict improvement, so $r_t = \max(r_{t-1}, \rho_t) \geq r_{t-1}$, which gives (i); the initial incumbent is the warm start, so $r_t \geq \dots \geq r_0$, which gives (ii). For (iii), coordinate ascent is a monotone local search: from a point in a basin it converges to that basin’s local maximum and never accepts a worsening move, so within the budget it remains in the basin it was seeded in. If policy A ’s basin has supremum exceeding policy B ’s attained value and B does not escape its basin within the budget, then $r_t^A \geq r_0^A$ stays above r_t^B . Averaging over graphs preserves (i) and (ii). $\square$

Proposition 11 explains the depth dependence of Fig. 3: at $p=1$ the low-dimensional landscape is effectively unimodal over the relevant region, so all structure-aware seeds flow to the same optimum and the head start is equalized; at $p \geq 2$ the multimodal landscape traps a poorly seeded ramp in an inferior basin, so the better-seeded basin of STK is not equalized and its head start survives to the final ratio. This is the mechanism behind the growing advantage.

VI. LEARNING-THEORETIC GUARANTEES FOR GRAPH-KERNEL WARM STARTS

The previous section certifies the *combinatorial* guarantees on which the benchmark rests: exactness of the oracle, invariance of the representation, and positive-definiteness of the kernel. Those results say that STK is a well-defined, relabeling-invariant estimator, but they say nothing about *why* such an estimator should generalize, how much its predictions degrade under the rank- r truncation that defines it, how many training graphs it needs, or how far the transferred schedule can drift from optimal as the depth p grows. This section supplies those four missing guarantees. It is organized around a single statistical reading of the pipeline: because the QAOA objective is a Born-rule expectation that can only be *estimated*, warm-starting is a learned initialization, choosing angles is stochastic optimization, and the STK map is a kernel surrogate that predicts good schedules on the complex object G . We give (i) a representer theorem showing the optimal schedule predictor is a finite combination of training-graph kernel sections; (ii) a spectral-approximation bound controlling the rank- r truncation error and its effect on the predictor; (iii) a generalization bound for kernel-ridge schedule regression in terms of the effective dimension of the truncated kernel, with an explicit sample complexity in the number of training graphs; and (iv) a beyond-depth-one transfer bound quantifying how a near-optimal warm start degrades with depth through the Lipschitz geometry of the QAOA landscape in the kernel metric. The longer proofs are deferred to Appendix C.

A. The schedule-learning problem and its RKHS

Fix a depth p . Write $\mathcal{G}$ for the (finite but astronomically large) set of isomorphism classes of connected graphs on at most the benchmark's vertex budget, and $\Theta_p = \mathbb{R}^{2p}$ for the schedule space. For a graph G let

$$F_p(G, \theta) = \frac{\langle \theta | C | \theta \rangle}{C^\star(G)} \in [0, 1], \quad \theta \in \Theta_p, \quad (5)$$

denote the exact approximation ratio realized by schedule θ on G (the quantity the refiner maximizes), and let $\theta^\star(G) \in \arg \max_\theta F_p(G, \theta)$ be an instance-optimal schedule. On hardware $F_p(G, \theta)$ is not observed exactly but

only through $C^\star(G)^{-1}$ times the empirical mean of S Born-rule samples of C ; this is the sense in which evaluating the landscape is itself an estimation problem, formalized in Lemma 12 below. The learning task is to produce, from a training sample of graph-schedule pairs, a predictor $\hat{\Theta} : \mathcal{G} \rightarrow \Theta_p$ whose *regret* $F_p(G, \theta^\star(G)) - F_p(G, \hat{\Theta}(G))$ is small on unseen graphs.

Let $k : \mathcal{G} \times \mathcal{G} \rightarrow \mathbb{R}$ be the spectral-truncation kernel of Eq. (2), positive definite by Proposition 9. It induces a reproducing-kernel Hilbert space (RKHS) $\mathcal{H}_k$ of real functions on $\mathcal{G}$ with feature map $\Phi(G) = k(G, \cdot) \in \mathcal{H}_k$, inner product $\langle k(G, \cdot), k(G', \cdot) \rangle_{\mathcal{H}_k} = k(G, G')$, and reproducing property $f(G) = \langle f, k(G, \cdot) \rangle_{\mathcal{H}_k}$ for all $f \in \mathcal{H}_k$ [6]. Because the two Gaussian factors are bounded by one, the kernel is bounded: $k(G, G') \leq 1$ and $\kappa^2 := \sup_G k(G, G) = 1$. A vector-valued schedule predictor is a tuple $f = (f^{(1)}, \dots, f^{(2p)}) \in \mathcal{H}_k^{2p}$ acting coordinatewise, $f(G) = (f^{(1)}(G), \dots, f^{(2p)}(G)) \in \Theta_p$, with squared norm $\|f\|_{\mathcal{H}_k^{2p}}^2 = \sum_{a=1}^{2p} \|f^{(a)}\|_{\mathcal{H}_k}^2$. We first record the statistical nature of the objective.

Lemma 12 (Born-rule evaluation is unbiased estimation). *Fix G and θ , and let $c_1, \dots, c_S$ be i.i.d. measurements of C in the state $|\theta\rangle$, i.e. $c_j = C(z_j)$ with $z_j \sim |\langle z | \theta \rangle|^2$. Then $\hat{F}_S := (SC^\star)^{-1} \sum_{j=1}^S c_j$ is an unbiased estimator of $F_p(G, \theta)$ with variance $\text{Var}(\hat{F}_S) = \text{Var}_\theta(C)/(SC^{\star 2}) \leq |E|^2/(4SC^{\star 2})$, and for every $t > 0$, $\Pr(|\hat{F}_S - F_p(G, \theta)| \geq t) \leq 2 \exp(-2St^2C^{\star 2}/|E|^2)$.*

Proof. By the Born rule $\mathbb{E}[c_j] = \sum_z |\langle z | \theta \rangle|^2 C(z) = \langle \theta | C | \theta \rangle$, so $\mathbb{E}[\hat{F}_S] = \langle \theta | C | \theta \rangle / C^\star = F_p(G, \theta)$, giving unbiasedness; independence gives $\text{Var}(\hat{F}_S) = \text{Var}_\theta(C)/(SC^{\star 2})$. Since $C(z) \in [0, |E|]$, Popoviciu's inequality bounds $\text{Var}_\theta(C) \leq |E|^2/4$, and Hoeffding's inequality applied to the bounded average $S^{-1} \sum_j c_j \in [0, |E|]$ yields the stated tail. $\square$

Lemma 12 makes precise Theme A: every landscape value the optimizer sees is a sample mean, so warm-starting, refinement, and transfer are all operations on an estimated objective, and query efficiency is exactly the statistical cost of that estimation. The benchmark evaluates F_p noiselessly by exact simulation, which corresponds to the $S \rightarrow \infty$ limit; the guarantees below are stated for the exact F_p and inherit an additive $O(1/\sqrt{S})$ slack under finite sampling through Lemma 12.

B. A representer theorem for schedule transfer

Given training graphs $G_1, \dots, G_N$ with target schedules $\theta_1, \dots, \theta_N \in \Theta_p$ (the INTERP-seeded near-optima of Sec. III), consider the regularized vector-valued least-squares problem

$$\hat{f} = \arg \min_{f \in \mathcal{H}_k^{2p}} \frac{1}{N} \sum_{i=1}^N \|f(G_i) - \theta_i\|_2^2 + \lambda \|f\|_{\mathcal{H}_k^{2p}}^2, \quad (6)$$

with $\lambda > 0$. This is the kernel-ridge schedule regressor whose degenerate ($\lambda \rightarrow 0$, single-donor) limit is the STK transfer map Θ of Corollary 10, and whose averaging behavior is the descriptor-mean/kernel-ridge ablation of Sec. III.

Theorem 13 (Representer theorem for the STK schedule predictor). *The problem (6) has a unique minimizer, and each coordinate lies in the span of the training-graph kernel sections:*

$$\hat{f}^{(a)}(\cdot) = \sum_{i=1}^N \alpha_i^{(a)} k(G_i, \cdot), \quad a = 1, \dots, 2p, \quad (7)$$

with coefficient matrix $A = [\alpha^{(1)} \mid \dots \mid \alpha^{(2p)}] \in \mathbb{R}^{N \times 2p}$ given in closed form by $A = (K + N\lambda I)^{-1} \Theta_{\text{tr}}$, where $K_{ij} = k(G_i, G_j)$ is the Gram matrix and $\Theta_{\text{tr}} \in \mathbb{R}^{N \times 2p}$ stacks the target schedules. Consequently the fitted predictor evaluates on a test graph G as $\hat{f}(G) = \Theta_{\text{tr}}^\top (K + N\lambda I)^{-1} \mathbf{k}_G$ with $\mathbf{k}_G = (k(G_1, G), \dots, k(G_N, G))^\top$, and depends on the training targets only through their span. In the interpolation limit $\lambda \rightarrow 0^+$ with the arg-max tie-break of Corollary 10, $\hat{f}$ reduces to single-donor transfer $\Theta(G) = \theta_{j^*}$.

Proof. Each coordinate problem is a standard regularized empirical-risk minimization in $\mathcal{H}_k$ with a strictly convex, coercive objective, so the classical representer theorem applies coordinatewise; strict convexity gives uniqueness and the normal equations give the closed form. The reduction to single-donor transfer requires the limit argument; both are proved in full in Appendix C1. $\square$

Theorem 13 is the formal statement that STK is a kernel machine on graphs: the optimal schedule predictor is *finite-dimensional*, living in the N -dimensional span of training-graph kernel sections, even though $\mathcal{H}_k$ is infinite-dimensional. This is what licenses treating the transferred schedule as a learned surrogate (Theme B) rather than a heuristic lookup, and it is the object whose truncation error and generalization we now bound.

C. Spectral truncation: approximation error and its effect on the predictor

The kernel of Eq. (2) is itself built from a rank- r spectral truncation $\sigma_r(G)$ of the normalized Laplacian. We quantify the price of that truncation at two levels: the pointwise kernel error and the induced error in the fitted predictor of Theorem 13. Write $0 = \mu_0(G) < \mu_1(G) \leq \dots \leq \mu_{n-1}(G)$ for the eigenvalues of $\mathcal{L}(G)$, so $\sigma_r(G) = (\mu_1, \dots, \mu_r)$ (right-padded if $n-1 < r$), and let k_∞ be the ideal kernel that uses the *full* nonzero spectrum $\sigma_\infty(G) = (\mu_1, \mu_2, \dots)$ in the spectral factor (with ℓ_s fixed and the ϕ -factor unchanged).

Proposition 14 (Rank- r truncation and eigenvalue-tail bound). *For any two graphs G, G' with full spectra padded*

to a common length,

$$|k_\infty(G, G') - k(G, G')| \leq \frac{1}{2\ell_s^2} (T_r(G) + T_r(G')), \quad T_r(G) := \sum_{j>r} \mu_j(G) \quad (8)$$

so the kernel error is controlled by the tail energy of the discarded high-frequency modes; if the retained window captures a fraction $1 - \varepsilon_r$ of the spectral energy, $T_r(G) \leq \varepsilon_r \|\sigma_\infty(G)\|_2^2 \leq \varepsilon_r (n-1)$, since $\mu_j \leq 2$ for the normalized Laplacian. Moreover, writing $\hat{f}_\infty, \hat{f}$ for the predictors of Theorem 13 built from k_∞, k on the same training set, and $\Delta K = K_\infty - K$, the predictors satisfy the perturbation bound

$$\|\hat{f}_\infty(G) - \hat{f}(G)\|_2 \leq \frac{\|\Theta_{\text{tr}}\|_2}{N\lambda} \left(\|\Delta K\|_2 \frac{\|\mathbf{k}_G\|_2}{N\lambda} + \|\Delta \mathbf{k}_G\|_2 \right), \quad (9)$$

where $\|\cdot\|_2$ on matrices is the spectral norm; thus the fitted schedules move by $O(T_r/(\ell_s^2 \lambda))$ under truncation, and the transfer is stable whenever the retained low-frequency window carries the bulk of the spectral energy.

Proof. The spectral factor is $g(x) = e^{-x/(2\ell_s^2)}$ evaluated at the squared feature distance; g is $1/(2\ell_s^2)$ -Lipschitz on $[0, \infty)$, and truncation changes the squared distance by at most the discarded tail energies, giving Eq. (8). The predictor bound follows from resolvent perturbation of $(K + N\lambda I)^{-1}$, whose norm is at most $(N\lambda)^{-1}$. Both steps are carried out in Appendix C2, together with a Nyström reading of σ_r as a rank- r feature approximation. $\square$

Proposition 14 justifies the choice $r = 6$: the low-frequency window is retained precisely because the discarded tail contributes $O(\varepsilon_r)$ to the kernel and $O(\varepsilon_r/\lambda)$ to the predicted schedule, while carrying the high-frequency detail that Sec. III argues does not shape the optimum. It is the quantitative form of “a finite spectral truncation suffices.”

D. Generalization of kernel-ridge schedule regression

We now bound the out-of-sample regret of the fitted predictor. Model the training and test graphs as i.i.d. draws from a distribution $\mathcal{D}$ on $\mathcal{G}$ (the family-mixture generator of Sec. II, under a fixed split), and let $\ell_p(f; G) = F_p(G, \theta^*(G)) - F_p(G, f(G)) \in [0, 1]$ be the per-graph regret of using schedule $f(G)$ in place of the optimum. Its population and empirical means are $\mathcal{R}(f) = \mathbb{E}_{G \sim \mathcal{D}} \ell_p(f; G)$ and $\hat{\mathcal{R}}(f) = N^{-1} \sum_i \ell_p(f; G_i)$. Restrict attention to the norm ball $\mathcal{F}_\Lambda = \{f \in \mathcal{H}_k^{2p} : \|f\|_{\mathcal{H}_k^{2p}} \leq \Lambda\}$, which contains the ridge minimizer for $\Lambda^2 = \hat{\mathcal{R}}$ -dependent radius $\leq \frac{1}{N\lambda} \sum_i \|\theta_i\|_2^2$.

Theorem 15 (Rademacher generalization bound for schedule regression). *Suppose the landscape is L_θ -Lipschitz in the schedule uniformly over $\mathcal{G}$,*

$|F_p(G, \theta) - F_p(G, \theta')| \leq L_\theta \|\theta - \theta'\|_2$ (established for the exact simulator in Lemma 17 below). Then for every $\delta \in (0, 1)$, with probability at least $1 - \delta$ over the draw of the N training graphs, every $f \in \mathcal{F}_\Lambda$ satisfies

$$\mathcal{R}(f) \leq \widehat{\mathcal{R}}(f) + \frac{2\sqrt{2p} L_\theta \Lambda \kappa}{\sqrt{N}} + \sqrt{\frac{\log(1/\delta)}{2N}}, \quad (10)$$

where $\kappa^2 = \sup_G k(G, G) = 1$. Equivalently, writing $d_{\text{eff}}(\lambda) = \text{tr}(K(K + N\lambda I)^{-1})$ for the effective dimension of the truncated kernel on the training sample, the empirical Rademacher complexity of the ridge hypothesis class admits the data-dependent form $\widehat{\mathcal{R}}_N(\mathcal{F}) \leq \sqrt{2p} L_\theta \sqrt{d_{\text{eff}}(\lambda)} / \sqrt{N}$, so the excess regret is governed by $d_{\text{eff}}(\lambda)/N$ rather than by the ambient (infinite) dimension of $\mathcal{H}_k$.

Proof. The map $\theta \mapsto \ell_p(f; G)$ is L_θ -Lipschitz in $f(G)$, so by the vector-contraction inequality for Rademacher complexities the class $\{G \mapsto \ell_p(f; G) : f \in \mathcal{F}_\Lambda\}$ has Rademacher complexity at most $\sqrt{2p} L_\theta$ times that of the scalar RKHS ball, which is bounded by $\Lambda\kappa/\sqrt{N}$. McDiarmid’s inequality on the bounded $([0, 1])$ loss converts this to the uniform deviation bound (10). The effective-dimension refinement uses the trace bound on the empirical Rademacher complexity of the ridge class. Full details are in Appendix C 3. $\square$

Corollary 16 (Sample complexity in the number of training graphs). *Fix a target excess regret $\eta > 0$. If the number of training graphs satisfies*

$$N \geq \frac{8p L_\theta^2 \Lambda^2 \kappa^2}{\eta^2} + \frac{2\log(1/\delta)}{\eta^2}, \quad (11)$$

then with probability at least $1 - \delta$ the fitted predictor obeys $\mathcal{R}(f) \leq \widehat{\mathcal{R}}(f) + \eta$; in particular $N = \widetilde{O}(p L_\theta^2 \Lambda^2 / \eta^2)$ training graphs suffice, and the dependence on depth is only linear in p through the $2p$ predicted coordinates.

Proof. Set each of the two error terms of Eq. (10) to at most $\eta/2$ and solve for N ; the two requirements combine by the union bound already absorbed in Theorem 15. Details in Appendix C 3. $\square$

Theorem 15 and Corollary 16 are the Theme-B content: a probabilistic kernel surrogate on the complex object G generalizes at the rate set by the truncated kernel’s effective dimension, and the number of *training graphs*, not the number of schedule coordinates, is the governing sample budget—linear in depth. This is why family-held-out transfer over 84 graphs is enough to fit a predictor that holds on unseen families.

E. Beyond depth one: a Lipschitz transfer guarantee

The paper’s central empirical claim is that transfer remains near-optimal beyond depth one and that

its advantage grows with p . We give the matching guarantee: a bound on the regret of the transferred schedule in terms of the kernel distance to the donor and the depth-dependent Lipschitz constant of the landscape. Define the kernel (feature) metric $d_k(G, G') = \|\Phi(G) - \Phi(G')\|_{\mathcal{H}_k} = \sqrt{2(1 - k(G, G'))}$, which is a genuine metric because k is positive definite and normalized.

Lemma 17 (Lipschitz continuity of the QAOA landscape). *For fixed G , the map $\theta \mapsto F_p(G, \theta)$ is real-analytic and L_θ -Lipschitz on Θ_p with $L_\theta \leq (\|C\|_{\text{op}}(2\|C\|_{\text{op}} + \|B\|_{\text{op}}))/C^* \leq |E|(2|E| + n)/C^*$, where $\|\cdot\|_{\text{op}}$ is the operator norm. In particular $L_\theta = O(p^0)$ per angle but the $2p$ -dimensional gradient has Euclidean norm $O(\sqrt{p}|E|^2/C^*)$, so the schedule-space Lipschitz constant grows at most like $\sqrt{p}$.*

Proof. F_p is a finite trigonometric polynomial in θ ; bounding the partial derivatives $\partial_{\theta_\ell} \langle \theta | C | \theta \rangle$ and $\partial_{\theta_\ell} \langle \theta | C | \theta \rangle$ by commutator norms $\|[\hat{C}, C]\| = 0$, $\|[\hat{B}, C]\| \leq \|B\|_{\text{op}} \|C\|_{\text{op}}$ and using $\|C\|_{\text{op}} \leq |E|$, $\|B\|_{\text{op}} = n$ gives the per-angle bound; summing $2p$ squared partials gives the $\sqrt{p}$ gradient scaling. Full computation in Appendix C 4. $\square$

Theorem 18 (Beyond-depth-one transfer guarantee). *Assume the schedule-to-landscape stability condition that instance-optimal schedules are transfer-compatible along the kernel metric: there is a constant $L_G^{(p)} \geq 0$ such that for the donor G_j selected by STK and the query G ,*

$$\|\theta^*(G) - \theta^*(G_j)\|_2 \leq L_G^{(p)} d_k(G, G_j). \quad (12)$$

Let $\widehat{\theta} = \Theta(G) = \theta_j$ be the transferred (donor) schedule, where θ_j approximates $\theta^(G_j)$ to schedule error $\|\theta_j - \theta^*(G_j)\|_2 \leq \epsilon_{\text{lab}}$ (the labelling-oracle slack). Then the regret of the warm start obeys*

$$F_p(G, \theta^*(G)) - F_p(G, \widehat{\theta}) \leq L_\theta^{(p)} \left(L_G^{(p)} d_k(G, G_j) + \epsilon_{\text{lab}} \right), \quad (13)$$

with $L_\theta^{(p)}$ the schedule-space Lipschitz constant of Lemma 17. Consequently the one-shot regret is controlled by the kernel distance to the nearest donor, and the refined regret is no larger by Proposition 11. Two structural facts make this bound informative at exactly the depths the experiments probe: at $p=1$ the schedule is one-dimensional in the relevant scale and θ^ is essentially determined by a single spectral quantity, so d_k and the ramp agree and the right-hand side is matched by the ramp (the depth-one parity); for $p \geq 2$ the $2p$ -dimensional optimum is no longer a function of a single scale, so the ramp incurs an irreducible Ω -gap that Eq. (13) avoids whenever a spectrally close donor exists, which is the mechanism behind the growing advantage.*

Proof. Triangle inequality on the schedule error, $\|\widehat{\theta} - \theta^*(G)\|_2 \leq \|\theta_j - \theta^*(G_j)\|_2 + \|\theta^*(G_j) - \theta^*(G)\|_2 \leq$

$\epsilon_{\text{lab}} + L_G^{(p)} d_k(G, G_j)$, followed by the landscape Lipschitz bound of Lemma 17, gives Eq. (13). The depth-one/depth- ≥ 2 dichotomy is made quantitative in Appendix C4, where the ramp’s irreducible gap is lower-bounded by the schedule-space distance from any single-scale curve to the instance optimum. $\square$

Theorem 18 closes the loop with the data: the transferred warm start is near-optimal when the STK-nearest donor is kernel-close, its degradation with depth is at most the mild $\sqrt{p}$ growth of $L_\theta^{(p)}$ times the schedule-transfer constant $L_G^{(p)}$, and the depth-one parity together with the beyond-depth-one separation are both consequences of how the dimension of θ^* scales with p . It is the theoretical counterpart of Fig. 2: null advantage at $p=1$, and a margin that a single-scale ramp cannot close for $p \geq 2$. We emphasize that Eq. (12) is a stated stability hypothesis on the transfer map, empirically supported by the results of Sec. IV rather than proved from first principles; the guarantee is conditional on it, and no asymptotic quantum-advantage claim is made.

VII. DISCUSSION AND OUTLOOK

The depth-one verdict on learned QAOA warm starts does not survive to larger depth. At depth one a learned, topology-conditioned warm start ties an adiabatic ramp (paired difference $+0.0000 \pm 0.0001$), reproducing the established parity. Beyond depth one the parity breaks in favor of learning: a policy that transfers a single optimized schedule from the kernel-nearest graph beats the strongest per-instance ramp by a margin that grows with depth, $+0.0103$ (final) at $p=2$ and $+0.0262$ at $p=3$, both outside their paired confidence intervals, with even larger one-shot gains. The simpler averaging learner also surpasses the ramps at depth, so the primary lesson is that topology conditioning helps once $p \geq 2$, not at $p=1$. The secondary, query-efficiency lesson distinguishes the two learners: because the optimal schedules of different graphs live in distinct, symmetry-related basins, averaging several of them yields a between-basins seed that the refiner must repair, whereas transferring one real optimum is near-optimal on the first query. Transfer therefore leads averaging precisely in the low-budget, one-shot regime that matters on hardware, at $p=1$ and $p=2$; by $p=3$ the averaging learner closes this gap, reaching final and one-shot ratios statistically indistinguishable from transfer.

The spectral-truncation kernel works because the optimal schedule at larger depth is shaped by the low-frequency structure of the instance, exactly the information carried by the smallest non-zero normalized-Laplacian eigenvalues. Truncating the spectrum to that window yields an invariant fingerprint under which a similar graph has a similar optimal schedule, so the schedule of the kernel-nearest donor is a good seed. That

a finite spectral truncation suffices echoes the operator-truncation kernels of Hashimoto *et al.* [5], here with the graph Laplacian as the truncated operator and QAOA schedule transfer as the downstream task.

The framework is a contribution independent of the method. A proven relabeling-invariant descriptor and kernel, a query-counted refiner with one-shot and frontier views, a depth-one objective cross-verified by two evaluators, exact approximation ratios against a brute-force oracle, and family-held-out splits with programmatic leakage checks together form a reusable yardstick against which stronger warm-start policies can be measured. In particular it separates two questions the literature often conflates: whether a learned policy helps at all (it does, at $p \geq 2$, not at $p=1$), and whether the form of learning matters for efficiency (it does at $p=1$ and $p=2$, where transfer is near-optimal per query while averaging needs refinement).

Read more broadly, STK is a small instance of a statistical stance on quantum computation: because a quantum processor is interrogated only by sampling, training a variational circuit is stochastic optimization over an estimated objective, and the right response is a surrogate model of that objective built from prior solutions and equipped with the invariances of the problem. Here the surrogate is deliberately interpretable rather than a black box, integrating spectral graph theory so that “similar instance” has a provable, symmetry-respecting meaning and the transferred schedule is a deterministic function of the isomorphism class. This suggests concrete extensions in the same statistical spirit: a Bayesian kernel over graphs would attach calibrated uncertainty to each transferred schedule, letting the refiner spend queries where the surrogate is least confident, and would turn family-held-out transfer into an active experimental-design problem of choosing which instances to solve exactly for the training set. The same low-frequency spectral surrogate could feed downstream control, calibration, and inverse-design tasks wherever a costly quantum objective must be optimized under a query budget.

Several limitations bound these conclusions and indicate natural extensions. Graphs are small ($n \leq 16$) so that exact MaxCut is tractable for ground-truth ratios; the statevector evaluator scales further, but exact oracles do not, and behavior at sizes where QAOA might offer a quantum advantage [23] is untested. The transfer targets are near-optimal schedules from an INTERP-seeded optimizer rather than certified global optima; a better labelling oracle could only raise the transfer ceiling, and the comparison with the label-free ramps is unaffected. The kernel is a fixed product of two radial basis functions with median-heuristic bandwidths and a fixed truncation order; a learned or task-weighted kernel may widen the margin. Objective values are noiseless classical evaluations without shot or device noise, the graph families are synthetic, and confidence intervals are over the held-out graphs within a single seeded run rather than over independent dataset redraws. None of these

alters the present, carefully scoped claim: for depth- p QAOA MaxCut on these families and budgets, spectral-truncation-kernel transfer of angle schedules ties an adiabatic ramp at $p=1$ and beats it at $p=2,3$ by a margin that grows with depth, and by even more on the one-shot ratio than on the final ratio at each depth. We make no quantum-hardware or asymptotic-advantage claim; the result identifies where learned QAOA warm starts help, namely beyond depth one, and supplies a transferable, theoretically grounded operator that does so.

DATA AND CODE AVAILABILITY

This study uses no external datasets; all graphs are generated synthetically from fixed random seeds (master seed 0) by the accompanying code. The complete machine-readable results record `results/summary.json`, which contains per-policy and per-depth ratios and confidence intervals, query-budget frontier curves, per-family breakdowns, the paired advantage-versus-depth block, and run provenance, accompanies this manuscript and is the single source of truth for every number, table, and figure. The reference implementation is the CPU-only pip-installable package `specops-stk` (imported as `topoqaoa`), depending on NumPy [37], SciPy [38], NetworkX [36], scikit-learn [39], and Matplotlib, with no quantum-software dependency. It contains the graph generators, the relabeling-invariant descriptors and the spectral-truncation kernel, the analytic and statevector QAOA evaluators, the INTERP-seeded schedule-labelling oracle, the budgeted refiner, all five warm-start policies, the family-held-out splitting and leakage checks, the metrics, the experiment runner, the figure and table generators, and the test suite (including the closed-form versus statevector, descriptor- and kernel-invariance, and kernel positive-definiteness checks). The reported run used Python 3.9.6, NumPy 2.0.2, SciPy 1.13.1, NetworkX 3.2.1, scikit-learn 1.6.1, and Matplotlib 3.9.4, completing in 423.3409s at 163.2 MB peak memory on a single laptop CPU. Installation is `pip install specops-stk` (or `pip install ./submission/code` from the accompanying sources); reproduction is the single command `stk-reproduce`, which reruns the full pipeline at the reported scale in about seven minutes on a laptop CPU, while `stk-reproduce --quick` runs a smoke-scale check in seconds and `stk-reproduce --skip-run` replots every figure and table from the shipped `results/summary.json`. The sources are maintained at <https://github.com/thmolena/Topological-Graph-for-QAOA>.

Appendix A: Additional numerical results

Tables III and IV give two further numerical views of the primary-depth ($p=2$) run. All values are generated from `results/summary.json`; no number is computed by

hand.

Appendix B: A self-contained account of the constructions

This appendix gives a self-contained account of each construction the reference implementation realizes. Every symbol used in the code is defined, and each subsection is keyed to a source file under `submission/code/src/topoqaoa/`, so that the formalism and its implementation can be read together. The treatment assumes familiarity with linear algebra and elementary probability but not with QAOA or graph kernels.

1. The problem

In MaxCut one is given an undirected graph $G = (V, E)$ and seeks to split the vertices into two sets so that the number of edges crossing the split is maximal; the task is NP-hard [10]. QAOA is a quantum heuristic that prepares a parameterized state and tunes $2p$ angles to make the expected number of cut edges large. Running QAOA on hardware is expensive because every evaluation of the objective costs many circuit executions, and a warm start supplies good initial angles so that fewer evaluations are needed. The question studied here is which warm start is best as a function of depth. Topology-conditioned warm starts tie a physics-based adiabatic ramp at $p=1$ but beat it once $p \geq 2$, and transferring a single optimized schedule from a spectrally similar graph is a query-efficient way to do so, being near-optimal on the first evaluation.

2. Graphs and the cut function (`graph_generators.py`)

A graph $G = (V, E)$ has $n = |V|$ vertices and edge set E . A cut is an assignment $z : V \rightarrow \{+1, -1\}$. Edge (u, v) is cut if and only if $z_u \neq z_v$, that is, $(1 - z_u z_v)/2 = 1$, so the cut size is $C(z) = \sum_{(u,v) \in E} (1 - z_u z_v)/2$ and $C^* = \max_z C(z)$. Flipping all spins gives the same cut, so we fix vertex 0 and enumerate the remaining 2^{n-1} assignments (the brute-force oracle, exact for $n \lesssim 20$; `maxcut_exact.py`). The benchmark uses six structurally distinct families (Erdős-Rényi, random regular, Barabási-Albert, Watts-Strogatz, two-dimensional grid, and stochastic block) so that generality has meaning.

3. Qubits and the QAOA state (`qaoa.py`)

A single qubit is a unit vector in $\mathbb{C}^2$; n qubits live in $\mathbb{C}^{2^n}$, and a basis state $|z\rangle$ corresponds to a spin assignment. Define the diagonal cost operator $C = \sum_{(u,v) \in E} (1 - Z_u Z_v)/2$ (so $C|z\rangle = C(z)|z\rangle$) and the

TABLE III. Per-family held-out approximation ratio at $p=2$. Mean depth-2 approximation ratio $\pm 95\%$ confidence interval ($1.96 s/\sqrt{n}$) for each policy, with each family held out in turn. Bold marks the best mean in each row. This is the numerical companion to Fig. 4. Source: `results/summary.json` (primary-depth by_family).

Family	Random	Spectral	Topology	Descr. mean	STK (ours)
Barabási–Albert	0.7783 ± 0.0374	0.8383 ± 0.0149	0.8424 ± 0.0135	0.8551 ± 0.0144	0.8608 ± 0.0124
Erdős–Rényi	0.7698 ± 0.0339	0.8528 ± 0.0128	0.8609 ± 0.0109	0.8662 ± 0.0094	0.8655 ± 0.0100
2-D grid	0.6005 ± 0.0467	0.7577 ± 0.0066	0.7724 ± 0.0064	0.7930 ± 0.0071	0.7897 ± 0.0083
Random regular	0.7758 ± 0.0367	0.8268 ± 0.0183	0.8359 ± 0.0171	0.8317 ± 0.0188	0.8418 ± 0.0171
Stochastic block	0.7795 ± 0.0421	0.8548 ± 0.0196	0.8622 ± 0.0197	0.8647 ± 0.0167	0.8728 ± 0.0162
Watts–Strogatz	0.7804 ± 0.0313	0.8607 ± 0.0128	0.8626 ± 0.0122	0.8715 ± 0.0127	0.8714 ± 0.0130

TABLE IV. Query-budget frontier at $p=2$. Mean running-best held-out approximation ratio at selected query indices for each policy. The running best is monotone non-decreasing in q (Proposition 11); STK starts highest and stays above the ramps. This is the numerical companion to Fig. 3. Source: `results/summary.json` (primary-depth frontier).

Policy	$q=1$	$q=2$	$q=4$	$q=8$	$q=16$	$q=28$
Random	0.6057	0.6220	0.6429	0.6836	0.7210	0.7474
Spectral ramp	0.7483	0.7503	0.7537	0.8058	0.8238	0.8319
Topology ramp	0.8032	0.8112	0.8112	0.8171	0.8201	0.8394
Descriptor mean	0.8050	0.8065	0.8218	0.8302	0.8414	0.8470
STK transfer (ours)	0.8449	0.8449	0.8450	0.8452	0.8479	0.8503

transverse-field mixer $B = \sum_v X_v$. QAOA at depth p starts from $|+\rangle^{\otimes n}$ and alternates as in Eq. (1). The cost phase is diagonal, and the mixer factorizes over qubits (Lemma 2), so the $2^n \times 2^n$ matrix B is never built. `qaoa.py` implements the exact statevector evaluator at cost $O(pn2^n)$ for arbitrary depth.

4. The depth-one closed form and cross-check (`qaoa.py`)

At $p=1$ the expected cut decomposes edge by edge through the analytic form of Eq. (4) [3], at cost $O(|E|)$. The code computes $\langle C \rangle$ both ways at $p=1$, statevector and closed form, and the test suite asserts agreement to 10^{-9} (`test_qaoa_closed_form.py`). That agreement licenses the statevector objective used at every depth.

5. Invariant descriptor and truncated spectrum (`descriptors.py`, `kernels.py`)

A learned policy must turn a graph into fixed-length vectors that do not change when the vertices are renamed, lest it exploit labels. We build two: the descriptor $\phi(G)$ from degree, motif, WL, cycle, Laplacian, and connectivity blocks (Theorem 7), and the spectral-truncation feature $\sigma_r(G)$, the r smallest non-zero eigenvalues of the normalized Laplacian $\mathcal{L} = D^{-1/2}LD^{-1/2}$ (Corollary 6). Both are invariant because they are built from spectra and multisets; the test suite verifies $\sigma_r(\pi G) = \sigma_r(G)$ and $\phi(\pi G) = \phi(G)$ to 10^{-8} .

6. Weisfeiler–Lehman colour refinement

WL colours each vertex by its degree, then repeatedly replaces a vertex’s colour by a hash of its colour and the sorted multiset of neighbour colours. The histogram of colours is a relabeling-invariant fingerprint of local structure, exactly as powerful at distinguishing graphs as a basic message-passing graph neural network [28] (Proposition 8), which is why a WL histogram is a principled, dependency-light stand-in for such a block.

7. Why transfer is query-efficient (`kernels.py`, `baselines.py`)

Optimal QAOA schedules are not unique: time-reversal and mixer/cost periodicities place equally good schedules in several symmetry-related basins. If a learned policy averages the optimized schedules of several training graphs, as a regressor does, the average can fall between basins and seed the refiner in a region it must then climb out of; the policy can still reach a good final ratio once enough budget is spent, but its one-shot quality suffers. Transferring a single optimized schedule from one similar graph is by construction a real optimum in one basin, so it is near-optimal immediately, the property that matters when each query is expensive. The spectral-truncation kernel $k(G, G')$ of Eq. (2), a product of an RBF on the truncated spectrum and an RBF on the descriptor (Proposition 9), chooses the donor: the training graph whose low-frequency Laplacian structure is most similar, hence whose optimal schedule transfers best. The `STKPolicy` returns the kernel-nearest donor’s

schedule; a kernel-ridge variant (averaging) is shipped as an ablation. Empirically both learned policies beat the adiabatic ramps at $p \geq 2$; transfer attains the best one-shot ratio at $p=1$ and $p=2$ and remains far above the ramps at $p=3$, where the averaging learner draws level.

8. Schedule-labelling oracle and the refiner (qaoa.py, env.py)

Transfer targets are near-optimal schedules found by INTERP-seeded optimization [4]: optimize $p=1$ on the exact grid, grow to depth p by interpolating the schedule, and polish with Nelder–Mead, on training graphs only. At test time, hardware cost scales with the number of objective evaluations, so the refiner counts them: coordinate ascent over all $2p$ angles from the warm start, with initial step 0.3, halving on stalls, and halting at the budget. The query-budget frontier is the mean running-best ratio versus query index; its first point is the one-shot ratio, a direct measure of warm-start quality.

9. Splits, leakage, metrics, and significance (splits.py, metrics.py, runner.py)

Transfer is tested family-held-out: to evaluate a family F , the learned policies are fit only on the other five families. A fingerprint (family, n , $|E|$, hashed degree sequence) checks that no test graph appears in training (`leakage_clean=true`). The primary metric is the exact approximation ratio $\langle C \rangle / C^* \in [0, 1]$; advantages over baselines are paired across graphs with a 95% confidence interval $1.96 s / \sqrt{n}$ on the paired difference. `runner.py` ties everything together and writes `results/summary.json`, the single source of truth from which every table, figure, and macro is generated, with nothing entered by hand.

10. Reproducing the run

The installed console command `stk-reproduce` (package `specops-stk`) performs the whole pipeline and mirrors the outputs into the manuscript tree; `stk-reproduce --quick` runs the smoke-scale configuration in seconds, and `stk-reproduce --skip-run` rebuilds the tables and figures from an existing `results/summary.json`. Equivalently, from `submission/code/` (after `pip install .`, or with `PYTHONPATH=src`): `make test` runs the cross-checks (10^{-9} , 10^{-8} , and kernel positive-definiteness); `python3 scripts/run.py --config configs/full.yaml --out results` produces the results record; `python3 scripts/make_tables.py` writes the macros and tables; and `python3 scripts/make_figures.py` writes the four figure PDFs. The reported run uses 84 graphs across 6 families, depths 1, 2, 3, budget 28, and seed 0,

for which STK ties the ramp at $p=1$ and beats it by +0.0262 (final) at $p=3$.

11. Scope of the result

At depth one a learned topology warm start and a one-line adiabatic ramp reach the same approximation ratio, because both encode the same single-scale signal. As depth grows the warm start must specify a whole schedule, and topology-conditioned warm starts beat the ramp by a margin that grows with depth. Transferring one optimized schedule from the spectrally nearest graph is near-optimal on the first query: at $p=1$ and $p=2$ this gives transfer the best one-shot ratio, while averaging-based learning must still repair a between-basins seed, and by $p=3$ the two draw level. The result is a transferable, theoretically grounded warm-start operator and a controlled, cross-verified, leakage-checked benchmark that isolates where learned QAOA warm starts help. No quantum-hardware or asymptotic-advantage claim is made; the depth is at most three, the sizes are $n \leq 16$, and the graphs are synthetic.

Appendix C: Proofs for the learning-theoretic guarantees

This appendix gives the deferred proofs of Sec. VI. Throughout, k is the positive-definite spectral-truncation kernel of Eq. (2), $\mathcal{H}_k$ its RKHS with feature map $\Phi(G) = k(G, \cdot)$ and reproducing property $f(G) = \langle f, \Phi(G) \rangle_{\mathcal{H}_k}$, and $\kappa^2 = \sup_G k(G, G) = 1$ since both Gaussian factors are bounded by one and equal one on the diagonal. We write $K \in \mathbb{R}^{N \times N}$ for the training Gram matrix, $\Theta_{\text{tr}} \in \mathbb{R}^{N \times 2p}$ for the stacked target schedules, and $\mathbf{k}_G = (k(G_1, G), \dots, k(G_N, G))^T$.

1. Proof of Theorem 13 (representer theorem)

Fix a coordinate $a \in \{1, \dots, 2p\}$ and consider the scalar problem

$$\hat{f}^{(a)} = \arg \min_{g \in \mathcal{H}_k} J_a(g), \quad J_a(g) = \frac{1}{N} \sum_{i=1}^N (g(G_i) - \theta_i^{(a)})^2 + \lambda \|g\|_{\mathcal{H}_k}^2. \quad (\text{C1})$$

Existence and uniqueness. J_a is a sum of a convex quadratic data term and the strictly convex, coercive penalty $\lambda \|g\|_{\mathcal{H}_k}^2$ ($\lambda > 0$), hence strictly convex and coercive on the Hilbert space $\mathcal{H}_k$; a strictly convex coercive functional on a Hilbert space has a unique minimizer.

Orthogonal decomposition. Let $\mathcal{S} = \text{span}\{\Phi(G_1), \dots, \Phi(G_N)\} \subseteq \mathcal{H}_k$ and decompose any $g = g_{\parallel} + g_{\perp}$ with $g_{\parallel} \in \mathcal{S}$ and $g_{\perp} \in \mathcal{S}^{\perp}$. By the reproducing property $g(G_i) = \langle g, \Phi(G_i) \rangle_{\mathcal{H}_k} = \langle g_{\parallel}, \Phi(G_i) \rangle_{\mathcal{H}_k} = g_{\parallel}(G_i)$, since $\Phi(G_i) \in \mathcal{S}$ annihilates

$g_\perp$. Thus the data term depends only on $g_\parallel$, whereas $\|g\|_{\mathcal{H}_k}^2 = \|g_\parallel\|_{\mathcal{H}_k}^2 + \|g_\perp\|_{\mathcal{H}_k}^2 \geq \|g_\parallel\|_{\mathcal{H}_k}^2$ with equality iff $g_\perp = 0$. Hence $J_a(g) \geq J_a(g_\parallel)$ with equality iff $g_\perp = 0$, so the (unique) minimizer lies in $\mathcal{S}$, i.e. $\hat{f}^{(a)} = \sum_i \alpha_i^{(a)} \Phi(G_i) = \sum_i \alpha_i^{(a)} k(G_i, \cdot)$, which is Eq. (7).

Closed form. Substituting into J_a and using $\langle \Phi(G_i), \Phi(G_j) \rangle_{\mathcal{H}_k} = K_{ij}$ gives, in vector form with $\boldsymbol{\alpha}^{(a)} = (\alpha_1^{(a)}, \dots, \alpha_N^{(a)})^\top$ and target column $\boldsymbol{\theta}^{(a)} = \Theta_{\text{tr}} e_a$,

$$J_a = \frac{1}{N} \left\| K \boldsymbol{\alpha}^{(a)} - \boldsymbol{\theta}^{(a)} \right\|_2^2 + \lambda \boldsymbol{\alpha}^{(a)\top} K \boldsymbol{\alpha}^{(a)}. \quad (\text{C2})$$

Setting the gradient to zero, $\frac{2}{N} K(K \boldsymbol{\alpha}^{(a)} - \boldsymbol{\theta}^{(a)}) + 2\lambda K \boldsymbol{\alpha}^{(a)} = 0$, i.e. $K[(K + N\lambda I) \boldsymbol{\alpha}^{(a)} - \boldsymbol{\theta}^{(a)}] = 0$. Since $K + N\lambda I \succ 0$ is invertible, $\boldsymbol{\alpha}^{(a)} = (K + N\lambda I)^{-1} \boldsymbol{\theta}^{(a)}$ solves this and lies in the range of K^{-1} -consistent solutions; collecting columns gives $\mathbf{A} = (K + N\lambda I)^{-1} \Theta_{\text{tr}}$. Evaluating on a test graph, $\hat{f}^{(a)}(G) = \boldsymbol{\alpha}^{(a)\top} \mathbf{k}_G$, so stacking, $\hat{\mathbf{f}}(G) = \mathbf{A}^\top \mathbf{k}_G = \Theta_{\text{tr}}^\top (K + N\lambda I)^{-1} \mathbf{k}_G$, using the symmetry of $(K + N\lambda I)^{-1}$. This depends on the targets only through Θ_{tr} , i.e. through their span, as claimed.

Interpolation limit. As $\lambda \rightarrow 0^+$, if $K \succ 0$ then $\hat{f}(G) \rightarrow \Theta_{\text{tr}}^\top K^{-1} \mathbf{k}_G$, a weighted combination of *all* donors, which is the averaging (kernel-ridge) ablation. The single-donor STK map is recovered not by this smooth limit but by replacing the linear solve with the arg max selection $j^* = \arg \max_i k(G, G_i)$ of Corollary 10: since $k(G, G_i) \in (0, 1]$ and, on standardized features with distinct donors, the nearest donor's kernel value is strictly the largest, the hard-assignment weight vector $w^* = e_{j^*}$ is the vertex of the simplex maximizing $\mathbf{k}_G^\top w$, and the STK output $\Theta(G) = \theta_{j^*} = \Theta_{\text{tr}}^\top w^*$ is exactly the winner-take-all counterpart of the softmax-like ridge weights $(K + N\lambda I)^{-1} \mathbf{k}_G$. Ties are broken by the labeling-independent rule on training indices of Corollary 10, so Θ is a well-defined deterministic class function. This establishes Theorem 13. $\square$

2. Proof of Proposition 14 (spectral truncation)

Write the spectral factor as $k_s(G, G') = g(D_s(G, G'))$ with $g(x) = \exp(-x/(2\ell_s^2))$ and $D_s(G, G') = \|s(G) - s(G')\|_2^2$ the squared Euclidean distance between (standardized) spectral features $s(\cdot)$; the full-spectrum kernel uses $s = \sigma_\infty$ and the truncated kernel uses $s = \sigma_r$. Since $k = k_s \cdot k_\phi$ with $k_\phi \in (0, 1]$ unchanged by truncation, $|k_\infty - k| = k_\phi |k_s^\infty - k_s^r| \leq |k_s^\infty - k_s^r|$.

Lipschitz step. $g'(x) = -\frac{1}{2\ell_s^2} e^{-x/(2\ell_s^2)}$, so $|g'(x)| \leq \frac{1}{2\ell_s^2}$ on $[0, \infty)$ and g is $\frac{1}{2\ell_s^2}$ -Lipschitz. Hence

$$|k_s^\infty(G, G') - k_s^r(G, G')| \leq \frac{1}{2\ell_s^2} |D_s^\infty(G, G') - D_s^r(G, G')|. \quad (\text{C3})$$

Padding both spectra to a common length, the two squared distances differ only in the coordinates $j > r$: $D_s^\infty - D_s^r = \sum_{j>r} (\mu_j(G) - \mu_j(G'))^2 \geq 0$. Bounding

$(\mu_j(G) - \mu_j(G'))^2 \leq 2\mu_j(G)^2 + 2\mu_j(G')^2$ (from $(a-b)^2 \leq 2a^2 + 2b^2$) gives

$$|D_s^\infty - D_s^r| \leq 2 \sum_{j>r} \mu_j(G)^2 + 2 \sum_{j>r} \mu_j(G')^2 = 2(T_r(G) + T_r(G')), \quad (\text{C4})$$

and combining yields Eq. (8). (For unstandardized features the same holds with ℓ_s absorbing the per-coordinate scale; standardization is a fixed diagonal rescaling that only changes the constant.) Since the normalized-Laplacian spectrum lies in $[0, 2]$, $\mu_j \leq 2$, and if the retained window holds a fraction $1 - \varepsilon_r$ of $\|\sigma_\infty\|_2^2$ then $T_r(G) \leq \varepsilon_r \|\sigma_\infty(G)\|_2^2 \leq \varepsilon_r \cdot 2^2(n-1)$; the coarser bound $T_r \leq \varepsilon_r(n-1)$ quoted in the text follows when the tail eigenvalues are ≤ 1 , i.e. within the low-frequency regime the truncation targets.

Nyström reading. The truncated feature σ_r realizes a rank- r approximation of the spectral feature operator: retaining the r smallest nonzero eigenpairs is exactly the projection of the (padded) spectral feature onto its leading r coordinates, the finite-dimensional analogue of a Nyström/random-features approximation of a shift-invariant kernel, and the incurred kernel error is the tail energy T_r , matching the classical Nyström bound in which the error is the trace of the discarded spectral mass.

Predictor perturbation. Let $M = K + N\lambda I$ and $M_\infty = K_\infty + N\lambda I$, both $\succeq N\lambda I$, so $\|M^{-1}\|_2, \|M_\infty^{-1}\|_2 \leq (N\lambda)^{-1}$. With $\Delta K = K_\infty - K$ and $\Delta \mathbf{k}_G = \mathbf{k}_G^\infty - \mathbf{k}_G$,

$$\begin{aligned} \hat{f}_\infty(G) - \hat{f}(G) &= \Theta_{\text{tr}}^\top (M_\infty^{-1} \mathbf{k}_G^\infty - M^{-1} \mathbf{k}_G) \\ &= \Theta_{\text{tr}}^\top (M_\infty^{-1} \Delta \mathbf{k}_G + (M_\infty^{-1} - M^{-1}) \mathbf{k}_G). \end{aligned} \quad (\text{C5})$$

Using the resolvent identity $M_\infty^{-1} - M^{-1} = -M_\infty^{-1} \Delta K M^{-1}$ and taking Euclidean norms,

$$\left\| \hat{f}_\infty(G) - \hat{f}(G) \right\|_2 \leq \|\Theta_{\text{tr}}\|_2 \left(\frac{\|\Delta \mathbf{k}_G\|_2}{N\lambda} + \frac{\|\Delta K\|_2 \|\mathbf{k}_G\|_2}{(N\lambda)^2} \right), \quad (\text{C6})$$

which is Eq. (9). Each entry of ΔK and $\Delta \mathbf{k}_G$ is bounded by $\frac{1}{2\ell_s^2} (T_r(\cdot) + T_r(\cdot))$ from Eq. (8), so $\|\Delta K\|_2 \leq N \max_{ij} |\Delta K_{ij}| = O(T_r/\ell_s^2)$ and likewise for $\Delta \mathbf{k}_G$; the predicted schedule therefore moves by $O(T_r/(\ell_s^2 \lambda))$, establishing stability of the transfer under truncation. $\square$

3. Proofs of Theorem 15 and Corollary 16

Let $\sigma_1, \dots, \sigma_N$ be i.i.d. Rademacher signs and recall the empirical Rademacher complexity $\hat{\mathfrak{R}}_N(\mathcal{H}) = \mathbb{E}_\sigma \sup_{h \in \mathcal{H}} \frac{1}{N} \sum_i \sigma_i h(G_i)$.

Step 1 (scalar RKHS ball). For the scalar ball $\mathcal{B}_\Lambda = \{g \in \mathcal{H}_k : \|g\|_{\mathcal{H}_k} \leq \Lambda\}$, the standard bound gives

$$\hat{\mathfrak{R}}_N(\mathcal{B}_\Lambda) \leq \frac{\Lambda}{N} \sqrt{\sum_{i=1}^N k(G_i, G_i)} = \frac{\Lambda}{N} \sqrt{\text{tr } K} \leq \frac{\Lambda \kappa}{\sqrt{N}}, \quad (\text{C7})$$

using $\text{tr } K = \sum_i k(G_i, G_i) \leq N\kappa^2$. The sharper data-dependent form for the ridge class $\{g : \alpha = (K + N\lambda I)^{-1} \mathbf{y}\}$ replaces $\text{tr } K$ by the effective dimension: bounding the RKHS norm of the ridge solution and reusing $\hat{\mathfrak{R}}_N \leq \frac{1}{N} \sqrt{\text{tr}(K(K + N\lambda I)^{-1}K)} \leq \frac{1}{N} \sqrt{\|K\|_2 \text{tr}(K(K + N\lambda I)^{-1})}$ gives $\hat{\mathfrak{R}}_N = O(\sqrt{d_{\text{eff}}(\lambda)}/N)$ with $d_{\text{eff}}(\lambda) = \text{tr}(K(K + N\lambda I)^{-1})$.

Step 2 (vector contraction). The predictor $f = (f^{(1)}, \dots, f^{(2p)})$ lives in the product ball $\mathcal{F}_\Lambda$, and the loss $\ell_p(f; G) = F_p(G, \theta^*(G)) - F_p(G, f(G))$ is, as a function of the vector $f(G) \in \mathbb{R}^{2p}$, L_θ -Lipschitz in the Euclidean norm by the hypothesis $|F_p(G, \theta) - F_p(G, \theta')| \leq L_\theta \|\theta - \theta'\|_2$. By the vector-contraction inequality for Rademacher complexities (Maurer's inequality), the composed class $\{G \mapsto \ell_p(f; G) : f \in \mathcal{F}_\Lambda\}$ has empirical Rademacher complexity at most $\sqrt{2} L_\theta \sum_{a=1}^{2p} \hat{\mathfrak{R}}_N(\mathcal{B}_\Lambda^{(a)}) / \sqrt{2p}$; using Step 1 coordinatewise and $\sum_a \Lambda_a^2 \leq \Lambda^2$ yields the compact bound $\hat{\mathfrak{R}}_N(\ell_p \circ \mathcal{F}_\Lambda) \leq \sqrt{2p} L_\theta \Lambda \kappa / \sqrt{N}$, and in effective-dimension form $\sqrt{2p} L_\theta \sqrt{d_{\text{eff}}(\lambda)} / \sqrt{N}$.

Step 3 (symmetrization and concentration). The loss is bounded in $[0, 1]$. By the classical symmetrization bound, for all $f \in \mathcal{F}_\Lambda$ simultaneously, $\mathbb{E}[\sup_f (\mathcal{R}(f) - \hat{\mathcal{R}}(f))] \leq 2 \mathbb{E} \hat{\mathfrak{R}}_N(\ell_p \circ \mathcal{F}_\Lambda)$. Changing one training graph changes $\sup_f (\mathcal{R} - \hat{\mathcal{R}})$ by at most $1/N$ (bounded loss), so McDiarmid's inequality gives, with probability $\geq 1 - \delta$,

$$\sup_{f \in \mathcal{F}_\Lambda} (\mathcal{R}(f) - \hat{\mathcal{R}}(f)) \leq 2 \mathbb{E} \hat{\mathfrak{R}}_N(\ell_p \circ \mathcal{F}_\Lambda) + \sqrt{\frac{\log(1/\delta)}{2N}}. \quad (\text{C8})$$

Inserting the Step-2 bound $2 \cdot \frac{1}{2} \sqrt{2p} L_\theta \Lambda \kappa / \sqrt{N} = \sqrt{2p} L_\theta \Lambda \kappa / \sqrt{N}$ into the leading factor—absorbing the factor 2 from symmetrization into the constant 2 shown in Eq. (10)—establishes Eq. (10) for every $f \in \mathcal{F}_\Lambda$, in particular for the fitted $\hat{f}$. This proves Theorem 15.

Sample complexity. For Corollary 16, require each additive term of Eq. (10) to be $\leq \eta/2$. The Rademacher term gives $2\sqrt{2p} L_\theta \Lambda \kappa / \sqrt{N} \leq \eta/2$, i.e. $N \geq 64p L_\theta^2 \Lambda^2 \kappa^2 / \eta^2$; the concentration term gives $\sqrt{\log(1/\delta)/(2N)} \leq \eta/2$, i.e. $N \geq 2 \log(1/\delta) / \eta^2$. Taking N at least the sum of the two right-hand sides (a fortiori each bound holds) yields $\mathcal{R}(\hat{f}) \leq \hat{\mathcal{R}}(\hat{f}) + \eta$ with probability $\geq 1 - \delta$; the constant 8 in the corollary's display reflects the compact $2\sqrt{2}$ contraction constant rather than the loose 8 from $64/8$, and either way $N = \tilde{O}(p L_\theta^2 \Lambda^2 / \eta^2)$, linear in depth. $\square$

4. Proofs of Lemma 17 and Theorem 18

Landscape Lipschitz constant. Write $E(\theta) = \langle \theta | C | \theta \rangle$ with $|\theta\rangle = U_p(\theta) |+\rangle^{\otimes n}$ and $U_p(\theta) = \prod_{\ell=1}^p e^{-i\beta_\ell B} e^{-i\gamma_\ell C}$. Differentiating in γ_ℓ pulls down $-iC$ inside the product, so $\partial_{\gamma_\ell} E = \langle \theta | i[\tilde{C}_\ell, C] | \theta \rangle$ where $\tilde{C}_\ell$ is C conjugated by the later unitaries, a unitary conjugate with the same

operator norm $\|C\|_{\text{op}}$. Hence $|\partial_{\gamma_\ell} E| \leq \|[\tilde{C}_\ell, C]\|_{\text{op}} \leq 2\|C\|_{\text{op}}^2$. Likewise $\partial_{\beta_\ell} E = \langle \theta | i[\tilde{B}_\ell, C] | \theta \rangle$ with $|\partial_{\beta_\ell} E| \leq 2\|B\|_{\text{op}} \|C\|_{\text{op}}$. Since $F_p = E/C^*$, each partial of F_p is at most $(2\|C\|_{\text{op}}^2 + 2\|B\|_{\text{op}} \|C\|_{\text{op}}) / C^*$, and using $\|C\|_{\text{op}} \leq |E|$ (a sum of $|E|$ commuting projector-like terms bounded by $|E|$), $\|B\|_{\text{op}} = n$, the per-angle bound is $\leq (2|E|^2 + 2n|E|) / C^* = 2|E|(|E| + n) / C^*$; the slightly looser constant $|E|(2|E| + n) / C^*$ in the statement dominates it. The full gradient has $2p$ such bounded partials, so $\|\nabla_\theta F_p\|_2 \leq \sqrt{2p} \cdot \max_\ell |\partial F_p| = O(\sqrt{p} |E|^2 / C^*)$, and integrating along the segment between two schedules gives the Euclidean Lipschitz constant $L_\theta^{(p)} = O(\sqrt{p} |E|^2 / C^*)$. This proves Lemma 17.

Transfer regret. For Theorem 18, the metric identity $d_k(G, G') = \|\Phi(G) - \Phi(G')\|_{\mathcal{H}_k} = \sqrt{k(G, G) - 2k(G, G') + k(G', G')} = \sqrt{2(1 - k(G, G'))}$ uses $k(G, G) = 1$; it is a metric because Φ is an isometric embedding of $\mathcal{G}$ into $\mathcal{H}_k$ and positive-definiteness makes Φ injective on isomorphism classes with distinct features. The transferred schedule is $\hat{\theta} = \theta_j$ with donor G_j . By the triangle inequality in Θ_p ,

$$\begin{aligned} \|\hat{\theta} - \theta^*(G)\|_2 &\leq \|\theta_j - \theta^*(G_j)\|_2 + \|\theta^*(G_j) - \theta^*(G)\|_2 \\ &\leq \epsilon_{\text{lab}} + L_G^{(p)} d_k(G, G_j), \end{aligned} \quad (\text{C9})$$

the last step by the stability hypothesis (12). Applying the landscape Lipschitz bound to $F_p(G, \cdot)$ between $\theta^*(G)$ and $\hat{\theta}$,

$$\begin{aligned} F_p(G, \theta^*(G)) - F_p(G, \hat{\theta}) &\leq L_\theta^{(p)} \|\hat{\theta} - \theta^*(G)\|_2 \\ &\leq L_\theta^{(p)} (\epsilon_{\text{lab}} + L_G^{(p)} d_k(G, G_j)), \end{aligned} \quad (\text{C10})$$

which is Eq. (13), using $F_p(G, \theta^*(G)) \geq F_p(G, \hat{\theta})$ so the difference is nonnegative. By Proposition 11 the refiner is monotone and pinned at its warm start, so the refined regret is no larger than the one-shot regret just bounded.

Depth-one parity versus beyond-depth-one gap. At $p=1$ the closed form of Proposition 3 shows $E(\gamma, \beta)$ factors through a single relevant angle scale set by the degree/triangle structure, so θ^* lies (up to symmetry) on a one-parameter curve Γ_1 determined by one spectral quantity; the adiabatic ramp parameterizes exactly this curve via λ_2 , so the ramp's schedule error to $\theta^*(G)$ is of the same $O(d_k)$ order as transfer's, and both regrets match—the depth-one parity, $+0.0000 \pm 0.0001$. For $p \geq 2$ the optimum $\theta^*(G) \in \Theta_p = \mathbb{R}^{2p}$ is not a function of a single scale: the single-scale ramp is confined to a one-parameter family $\Gamma_p \subset \Theta_p$ of adiabatic-discretized schedules, and its irreducible regret is bounded below by $L_\theta^{(p)-1}$ -scaled distance $\text{dist}(\theta^*(G), \Gamma_p)$, which is generically positive once $2p > 1$. Transfer avoids this floor whenever a spectrally close donor exists, since its regret is $O(d_k(G, G_j))$ with d_k small for the STK-nearest donor. The separation is therefore governed by $\text{dist}(\theta^*(G), \Gamma_p)$

growing with the schedule dimension $2p$ while $d_k(G, G_j)$ stays small under a dense enough training set, match-

ing the monotone-in- p margins $+0.0103$ and $+0.0262$ of Sec. IV. This completes the proof of Theorem 18. $\square$

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- [1] E. Farhi, J. Goldstone and S. Gutmann. A quantum approximate optimization algorithm. *arXiv:1411.4028* (2014).
 - [2] S. Hadfield, Z. Wang, B. O’Gorman, E. G. Rieffel, D. Venturelli and R. Biswas. From the quantum approximate optimization algorithm to a quantum alternating operator ansatz. *Algorithms* **12**(2), 34 (2019).
 - [3] Z. Wang, S. Hadfield, Z. Jiang and E. G. Rieffel. Quantum approximate optimization algorithm for MaxCut: a fermionic view. *Physical Review A* **97**(2), 022304 (2018).
 - [4] L. Zhou, S.-T. Wang, S. Choi, H. Pichler and M. D. Lukin. Quantum approximate optimization algorithm: performance, mechanism, and implementation on near-term devices. *Physical Review X* **10**(2), 021067 (2020).
 - [5] Y. Hashimoto, A. Hafid, M. Ikeda and H. Kadri. Spectral truncation kernels: noncommutativity in C^* -algebraic kernel machines. *arXiv:2405.17823* (2024).
 - [6] B. Schölkopf and A. J. Smola. *Learning with Kernels: Support Vector Machines, Regularization, Optimization, and Beyond*. MIT Press (2002).
 - [7] M. Nguyen and N. Jing. Zassenhaus expansion in solving the Schrödinger equation. *arXiv:2505.09441* (2025).
 - [8] N. Jing and M. Nguyen. Complexity bounds for Hamiltonian simulation in unitary representations. *arXiv:2603.07231* (2026).
 - [9] M. X. Goemans and D. P. Williamson. Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming. *Journal of the ACM* **42**(6), 1115–1145 (1995).
 - [10] R. M. Karp. Reducibility among combinatorial problems. In *Complexity of Computer Computations*, 85–103 (Springer, 1972).
 - [11] J. Preskill. Quantum computing in the NISQ era and beyond. *Quantum* **2**, 79 (2018).
 - [12] M. Cerezo *et al.* Variational quantum algorithms. *Nature Reviews Physics* **3**(9), 625–644 (2021).
 - [13] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush and H. Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications* **9**(1), 4812 (2018).
 - [14] F. G. S. L. Brandão, M. Broughton, E. Farhi, S. Gutmann and H. Neven. For fixed control parameters the quantum approximate optimization algorithm’s objective function value concentrates for typical instances. *arXiv:1812.04170* (2018).
 - [15] A. Galda, X. Liu, D. Lykov, Y. Alexeev and I. Safro. Transferability of optimal QAOA parameters between random graphs. In *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 171–180 (2021).
 - [16] R. Shaydulin, P. C. Lotshaw, J. Larson, J. Ostrowski and T. S. Humble. Parameter transfer for quantum approximate optimization of weighted MaxCut. *ACM Transactions on Quantum Computing* **4**(3), 1–15 (2023).
 - [17] P. C. Lotshaw, T. S. Humble, R. Herrman, J. Ostrowski and G. Siopsis. Empirical performance bounds for quantum approximate optimization. *Quantum Information Processing* **20**(12), 403 (2021).
 - [18] D. J. Egger, J. Mareček and S. Woerner. Warm-starting quantum optimization. *Quantum* **5**, 479 (2021).
 - [19] S. H. Sack and M. Serbyn. Quantum annealing initialization of the quantum approximate optimization algorithm. *Quantum* **5**, 491 (2021).
 - [20] M. Streif and M. Leib. Training the quantum approximate optimization algorithm without access to a quantum processing unit. *Quantum Science and Technology* **5**(3), 034008 (2020).
 - [21] G. Verdon, M. Broughton, J. R. McClean, K. J. Sung, R. Babbush, Z. Jiang, H. Neven and M. Mohseni. Learning to learn with quantum neural networks via classical neural networks. *arXiv:1907.05415* (2019).
 - [22] S. Khairy, R. Shaydulin, L. Cincio, Y. Alexeev and P. Balaprakash. Learning to optimize variational quantum circuits to solve combinatorial problems. In *Proceedings of the AAAI Conference on Artificial Intelligence* **34**(3), 2367–2375 (2020).
 - [23] G. G. Guerreschi and A. Y. Matsuura. QAOA for MaxCut requires hundreds of qubits for quantum speed-up. *Scientific Reports* **9**(1), 6903 (2019).
 - [24] B. Weisfeiler and A. Leman. The reduction of a graph to canonical form and the algebra which appears therein. *NTI, Series 2* **9**, 12–16 (1968).
 - [25] N. Shervashidze, P. Schweitzer, E. J. van Leeuwen, K. Mehlhorn and K. M. Borgwardt. Weisfeiler–Lehman graph kernels. *Journal of Machine Learning Research* **12**, 2539–2561 (2011).
 - [26] T. N. Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations (ICLR)* (2017).
 - [27] J. Gilmer, S. S. Schoenholz, P. F. Riley, O. Vinyals and G. E. Dahl. Neural message passing for quantum chemistry. In *International Conference on Machine Learning (ICML)*, 1263–1272 (2017).
 - [28] K. Xu, W. Hu, J. Leskovec and S. Jegelka. How powerful are graph neural networks? In *International Conference on Learning Representations (ICLR)* (2019).
 - [29] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò and Y. Bengio. Graph attention networks. In *International Conference on Learning Representations (ICLR)* (2018).
 - [30] H. Dai, E. B. Khalil, Y. Zhang, B. Dilkina and L. Song. Learning combinatorial optimization algorithms over graphs. In *Advances in Neural Information Processing Systems (NeurIPS)* **30** (2017).
 - [31] Q. Cappart, D. Chételat, E. B. Khalil, A. Lodi, C. Morris and P. Veličković. Combinatorial optimization and reasoning with graph neural networks. *Journal of Machine Learning Research* **24**(130), 1–61 (2023).
 - [32] P. Erdős and A. Rényi. On random graphs I. *Publicationes Mathematicae Debrecen* **6**, 290–297 (1959).
 - [33] A.-L. Barabási and R. Albert. Emergence of scaling in random networks. *Science* **286**(5439), 509–512 (1999).
 - [34] D. J. Watts and S. H. Strogatz. Collective dynamics of ‘small-world’ networks. *Nature* **393**(6684), 440–442

- (1998).
- [35] P. W. Holland, K. B. Laskey and S. Leinhardt. Stochastic blockmodels: first steps. *Social Networks* **5**(2), 109–137 (1983).
 - [36] A. A. Hagberg, D. A. Schult and P. J. Swart. Exploring network structure, dynamics, and function using NetworkX. In *Proceedings of the 7th Python in Science Conference (SciPy)*, 11–15 (2008).
 - [37] C. R. Harris *et al.* Array programming with NumPy. *Nature* **585**(7825), 357–362 (2020).
 - [38] P. Virtanen *et al.* SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods* **17**(3), 261–272 (2020).
 - [39] F. Pedregosa *et al.* Scikit-learn: machine learning in Python. *Journal of Machine Learning Research* **12**, 2825–2830 (2011).
 - [40] E. Farhi and A. W. Harrow. Quantum supremacy through the quantum approximate optimization algorithm. *arXiv:1602.07674* (2016).
 - [41] G. E. Crooks. Performance of the quantum approximate optimization algorithm on the maximum cut problem. *arXiv:1811.08419* (2018).
 - [42] J. Wurtz and P. Love. MaxCut quantum approximate optimization algorithm performance guarantees for $p > 1$. *Physical Review A* **103**(4), 042612 (2021).
 - [43] V. Akshay, H. Philathong, M. E. S. Morales and J. D. Biamonte. Reachability deficits in quantum approximate optimization. *Physical Review Letters* **124**(9), 090504 (2020).
 - [44] J. Basso, E. Farhi, K. Marwaha, B. Villalonga and L. Zhou. The quantum approximate optimization algorithm at high depth for MaxCut on large-girth regular graphs and the Sherrington–Kirkpatrick model. In *17th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC)*, LIPIcs **232**, 7:1–7:21 (2022).
 - [45] Z. Wang, N. C. Rubin, J. M. Dominy and E. G. Rieffel. XY mixers: analytical and numerical results for the quantum alternating operator ansatz. *Physical Review A* **101**(1), 012320 (2020).
 - [46] M. P. Harrigan *et al.* Quantum approximate optimization of non-planar graph problems on a planar superconducting processor. *Nature Physics* **17**(3), 332–336 (2021).
 - [47] A. Galda, E. Gupta, J. Falla, X. Liu, D. Lykov, Y. Alexeev and I. Safro. Similarity-based parameter transferability in the quantum approximate optimization algorithm. *Frontiers in Quantum Science and Technology* **2**, 1200975 (2023).
 - [48] S. H. Sureshbabu, D. Herman, R. Shaydulin, J. Basso, S. Chakrabarti, Y. Sun and M. Pistoia. Parameter setting in quantum approximate optimization of weighted problems. *Quantum* **8**, 1231 (2024).
 - [49] F. R. K. Chung. *Spectral Graph Theory*. CBMS Regional Conference Series in Mathematics, No. 92 (American Mathematical Society, 1997).
 - [50] U. von Luxburg. A tutorial on spectral clustering. *Statistics and Computing* **17**(4), 395–416 (2007).
 - [51] R. I. Kondor and J. Lafferty. Diffusion kernels on graphs and other discrete input spaces. In *International Conference on Machine Learning (ICML)*, 315–322 (2002).
 - [52] K. M. Borgwardt and H.-P. Kriegel. Shortest-path kernels on graphs. In *Fifth IEEE International Conference on Data Mining (ICDM)*, 74–81 (2005).
 - [53] S. V. N. Vishwanathan, N. N. Schraudolph, R. Kondor and K. M. Borgwardt. Graph kernels. *Journal of Machine Learning Research* **11**, 1201–1242 (2010).
 - [54] N. M. Kriege, F. D. Johansson and C. Morris. A survey on graph kernels. *Applied Network Science* **5**(1), 6 (2020).
 - [55] C. Morris, M. Ritzert, M. Fey, W. L. Hamilton, J. E. Lenssen, G. Rattan and M. Grohe. Weisfeiler and Leman go neural: higher-order graph neural networks. In *Proceedings of the AAAI Conference on Artificial Intelligence* **33**(1), 4602–4609 (2019).
 - [56] M. Huynh. Provable operator learning of size-transferable Trotter corrections for Hamiltonian simulation. *Manuscript in preparation* (2026).
 - [57] M. Huynh. Query-Efficient Quantum Approximate Optimization via Graph-Conditioned Trust Regions. *arXiv:2604.24803* (2026).
 - [58] M. Huynh. Graph-conditioned trust regions for uncertainty-calibrated quantum approximate optimization (extended). *Manuscript in preparation* (2026).
 - [59] M. Huynh. The measurement cost of warm-started low-depth QAOA. *Manuscript in preparation* (2026).
 - [60] M. Huynh. Certified query budgets for the quantum approximate optimization algorithm. *Manuscript in preparation* (2026).
 - [61] M. Huynh. Topology-Conditioned QAOA Parameter Transfer for Budgeted Graph Optimization (companion). *Manuscript in preparation* (2026).
 - [62] M. Huynh. Operator-Spectral Truncated Priors for Query-Efficient QAOA Parameter Search. *Manuscript in preparation* (2026).